

Proactive Transfer Admission Control for Emergency Departments

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Emergency Department (ED) crowding remains a persistent challenge, undermining timely care, elevating clinical risk, and placing substantial strain on hospital resources. While hospitals cannot easily control the arrival of walk-in emergency patients, they often have some flexibility in managing transfer admissions from other hospitals. Temporarily restricting such transfers during periods of high congestion can provide an effective operational lever to alleviate crowding. In this work, we investigate how predictive information, namely forecasts of future demand and inpatient admission delays, can be systematically incorporated into ED admission control decisions. We develop a queueing model that captures key patient flow dynamics, accounting for a time-varying operating environment, two distinct sources of ED arrivals, and delayed inpatient admissions (i.e., ED boarding). Building on this model, we derive the optimal admission control policy under a fluid approximation. The fluid-based policy can be seamlessly adapted to the stochastic system. It explicitly leverages predictive information, offers clear interpretability and practical implementability, and achieves performance close to the stochastic optimum. Through extensive numerical experiments and a case study based on real hospital data, we show that by proactively regulating the acceptance of transfer patients based on real-time congestion levels and anticipated future conditions, the proposed approach achieves a superior efficiency frontier balancing ED congestion and transfer acceptance than reactive admission policies. Our approach provides a systematic framework of integrating predictive analytics to proactively control admission that jointly addresses ED crowding and patient access.

Key words: emergency department, admission control, demand surge, queueing theory, asymptotic analysis

1. Introduction

Emergency department (ED) crowding has emerged as one of the most pervasive and persistent challenges in hospital care delivery. It reflects not only localized operational bottlenecks but also broader systemic strain, and has been shown to adversely affect patient outcomes and contribute to provider burnout (Hoot and Aronsky 2008, Sun et al. 2013). Despite widespread recognition of the problem, hospitals face substantial constraints in expanding capacity. Staffing shortages, rising costs, and the limited availability of physical space make it increasingly difficult for hospitals to simply “scale up” their way out of crowding (Janke and Venkatesh 2025).

Importantly, ED crowding is not merely an ED-level operational issue; it reflects a system-wide patient flow problem. A substantial portion of ED congestion arises from downstream bottlenecks,

particularly delays in admitting patients to inpatient units (McKenna et al. 2019). When inpatient beds are unavailable or admission processing is slow, admitted patients remain boarded in the ED, consuming space and staff resources and exacerbating crowding. Therefore, any meaningful approach to ED admission control must account not only for current ED conditions but also for inpatient-side congestion and the anticipated delays in admitting patients.

Given these barriers, there is growing interest in operational strategies that manage, not just accommodate, demand. While hospitals have little control over unscheduled “walk-in” emergency arrivals, many have some flexibility in how they manage inbound inter-facility transfer requests from one ED to another ED. Such ED-to-ED transfers play an important role in regionalized care systems, ensuring that patients receive the appropriate level of care and clinical services (Westfall 2024, Greenwood-Ericksen et al. 2025). Although the national percentage of ED patients in the United States who arrived from another ED is low (estimated at 1.1% in 2020-2022), it has increased over time (Peters et al. 2026) and can be substantially higher at some academic medical centers or referral centers. For example, Moore et al. (2023) study an ED where 13.6% of patients were received from another ED. Moreover, these encounter-based proportions do not capture transfer requests that are declined and therefore do not result in an ED visit. Transfer decisions are subject to applicable requirements, including those of the Emergency Medical Treatment and Labor Act (EMTALA) in the United States (Centers for Medicare and Medicaid Services 2026). Our focus is on academic medical centers and referral hospitals that receive transfer requests from community hospitals seeking care unavailable locally. For these receiving institutions, ED-to-ED transfers constitute a potentially modifiable stream of incoming patients whose timing and volume can meaningfully influence ED congestion (Greenwood-Ericksen et al. 2025). Temporarily regulating such admissions during periods of high crowding offers a promising lever for improving patient flow. In this work, we examine how to design effective admission control policies for ED-to-ED transfer patients (hereafter, transfer patients) to help alleviate ED crowding.

At the same time, recent advances in predictive analytics, combined with the growing availability of granular operational data, have created new opportunities for hospitals to anticipate changes in congestion rather than respond only to current conditions. In particular, forecasts of future ED demand and inpatient admission delays can provide information about how the operating environment is likely to evolve over the coming hours. This raises a natural question: how should such forecasts of time-varying system primitives be translated into transfer-acceptance decisions? In this work, we address this question by developing a forecast-informed admission-control framework that uses anticipated future demand and inpatient processing conditions to assess the congestion consequences of accepting transfer patients.

To study this problem, we develop a queueing model that captures the essential dynamics of ED patient flow relevant to transfer admission control. Our model distinguishes between two streams

of arrivals: (1) *Regular Arrivals*, including walk-ins, ambulance arrivals, and referrals from external providers; and (2) *Transfer Requests*, representing patients seeking transfer from other EDs. Admission control is applied exclusively to transfer requests. The model also explicitly captures delays in inpatient admission through a boarding stage, reflecting limited downstream capacity and other operational constraints. By accounting for the time-varying nature of ED operations and these key system features, the model provides a flexible framework for analyzing policies that balance patient waiting costs with the blocking costs associated with rejecting transfer requests.

For analytical development, we first study a deterministic fluid approximation in which the future trajectories of the system primitives are treated as given. This reference problem allows us to characterize how anticipated changes in demand, inpatient admission delays (inpatient processing), and staffing affect transfer-admission decisions. We derive the optimal fluid control and show that these future operating conditions enter the decision through a hitting-time-based congestion metric that summarizes the predicted duration of congestion. This structural characterization leads naturally to a forecast-informed policy for the stochastic system: at each review epoch, unknown future arrival and inpatient-processing rates are replaced by point forecasts constructed from currently available information. The resulting policy retains a simple threshold structure and is easy to implement. Under suitable regularity conditions, we further establish its asymptotic optimality in the many-server limit.

We then evaluate the policy through extensive simulation experiments and a case study based on real hospital data. The synthetic experiments consider different demand-surge patterns, review frequencies, and controlled forecast-error stress tests that vary the magnitude, persistence, horizon dependence, bias, and timing of forecast errors. These experiments allow us to identify which dimensions of forecast quality are most consequential for admission-control performance, highlighting the particular importance of accurately capturing the magnitude and timing of impending congestion. We then provide an end-to-end empirical implementation using data-driven models fitted to real hospital data to forecast ED arrival and inpatient processing rates. The results show that the forecast-informed policy achieves a more favorable trade-off between patient waiting and transfer rejection than commonly used reactive benchmarks.

Taken together, our work makes two primary contributions. First, we provide a structural analysis of transfer admission control in a nonstationary ED with downstream boarding. Using a fluid approximation, we show that the effects of anticipated future demand, inpatient processing, and staffing can be summarized through a congestion-clearing-time metric that determines the optimal admission threshold. While the resulting threshold structure is consistent with classical admission-control insights, our analysis identifies the underlying congestion externality generated by an additional transfer patients and establishes conditions under which the fluid-based policy

is optimal. We further provide an asymptotic optimality guarantee for its implementation in the stochastic system, thereby offering a principled foundation for translating forecasts of future operating conditions into admission decisions.

Secondly, we provide a comprehensive numerical and empirical evaluation of the proposed framework. The synthetic experiments isolate the effects of different forms of forecast error, while the hospital case study implements the policy using forecasting models trained and evaluated on real operational data. Across a broad range of operating conditions, the forecast-informed policy achieves a more favorable congestion-access trade-off than reactive admission rules. Together, these analyses demonstrate both the operational value of incorporating forecasts of future system conditions into transfer admission control and the importance of accurately capturing the magnitude and timing of impending congestion.

1.1. Literature review

Our work is mainly related to two lines of literature.

Admission control in queues Admission control has long been studied in queueing theory as a mechanism for mitigating congestion by balancing waiting costs against blocking or rejection penalties (see, e.g., Heyman 1968, Lewis 2001, Altman et al. 2002, Ormeci 2004, Koçağa and Ward 2010). In many classical stationary settings, the optimal admission policy exhibits a threshold structure (Naor 1969, Stidham Jr 2002, Ward and Kumar 2008). Admission control has also been studied in queueing networks with multiple stages or resources. For example, Silva et al. (2013) analyze admission control in a tandem loss system, although their model is time homogeneous.

Our theoretical model is more closely related to a separate literature on control of nonstationary queueing systems, in which time-varying arrival, service, or capacity rates are treated as primitives of the control problem. Bassamboo et al. (2005) study dynamic routing and admission control in a multiclass many-server system with temporally and stochastically varying arrival rates. Using a multi-scale fluid approximation, they derive asymptotically optimal controls based on the prevailing arrival-rate environment. Čudina and Ramanan (2011) develop a general framework for asymptotically optimal control of time-inhomogeneous queueing networks by first solving an associated fluid control problem. Zayas-Cabán and Lewis (2020) study admission control in loss systems with periodically varying parameters and characterize how optimal admission decisions adjust over predictable demand cycles. These studies establish an important precedent for treating temporal variation in system primitives explicitly and for using fluid or other approximations to obtain tractable control policies. In our reference model, the future trajectories of the primitive rates are treated as given, and the fluid analysis characterizes how their temporal evolution affects the optimal admission decision. The ED setting, however, introduces an additional source of

time variation through downstream inpatient processing. In particular, accepting a transfer patient affects congestion both directly (through the current ED workload) and indirectly (through future boarding dynamics). Our analysis shows that the combined effect of anticipated arrivals, inpatient processing, and staffing can be summarized through a congestion-clearing-time statistic that determines the admission threshold.

A distinct literature studies queueing control under richer forms of information about future stochastic events. Spencer et al. (2014) consider admission control when future arrival and service-event realizations are revealed over a look-ahead window. Motivated by ED diversion, Xu and Chan (2016) study proactive policies that exploit predictions of future arrival and service-token processes and analyze the effect of prediction noise. More recently, Peng (2024) develops a general forecast-information framework for admission control that encompasses a range of information structures, including no future information and full future information. Our information structure is deliberately coarser. We do not assume that the manager observes or predicts future sample-path realizations of individual arrivals, service completions, or inpatient-bed releases. Instead, with a view towards practical implementation, we replace the future primitive-rate trajectories in the reference model with point forecasts of aggregate arrival and inpatient-processing rates available at each decision epoch.

Relatedly, Bertsimas et al. (2022) operationalize machine-learning forecasts by embedding them into hospital information systems to support daily bed-placement decisions. More broadly, (Ata and Peng 2020, Delana et al. 2021, Ao et al. 2024) study how anticipatory information can be used to improve resource-allocation and service decisions in stochastic systems.

Managing Congestion in the ED The conceptual model of Asplin et al. (2003) frames ED crowding in three interdependent components: input, throughput, and output. Crowding is essentially an imbalance where input exceeds throughput and output capacity in the acute care system. Extensive work in the emergency medicine literature has documented the causes and consequences of ED crowding, including its effects on patient safety, service delays, and provider well-being (see, for example, Hoot and Aronsky 2008, Bernstein et al. 2009, Morley et al. 2018). Practices in ED-to-ED transfers and acceptance decisions have also been studied (Greenwood-Ericksen et al. 2025).

Within operations research, ED congestion has been studied from multiple angles using stochastic modeling, empirical analysis, and simulation. Queueing models have been widely applied to support operational decisions. For example, Green et al. (2006) and Yom-Tov and Mandelbaum (2014) use queueing analysis to determine appropriate staffing levels under varying demand conditions. Chan et al. (2022) study a fluid control problem for nurse reassignment in an ED. Jacobson et al. (2012) and Saghaian et al. (2012) show that streaming and prioritization schemes can be effective tools for improving system performance, particularly when the ED experiences extreme levels of

crowding. Shi et al. (2016) develop a stochastic network model of inpatient operations and show that inpatient discharge policies can have a substantial impact on ED boarding times. A parallel empirical literature investigates how ED clinicians adapt their behavior in response to congestion. For example, Song et al. (2015) analyze how alternative queue configurations influence ED physician productivity. Batt and Terwiesch (2017) document the phenomenon of early task initiation, where upstream staff (e.g., triage nurses) begin tasks typically performed by downstream providers when the ED becomes busy. Ding et al. (2019) and Li et al. (2021) show that prioritization decisions also shift with crowding. Adepoju et al. (2023) studied how reactive hospital-wide responses to ED crowding may have unintended negative consequences. Recently, Canellas et al. (2025) consider fair patient prioritization and placement in ED operations, using a predictive-prescriptive mixed integer linear program that breaks predicted ED length of stay into actionable components. Our work contributes to this literature by examining a complementary and understudied operational lever: admission control for inter-facility transfer patients.

1.2. Notations

For integer $n \geq 1$, we denote $[n] = \{1, 2, \dots, n\}$ as the set of integers from 1 to n . For $x \in \mathbb{R}$, denote $x_+ = \max\{x, 0\}$. For set S , denote $|S|$ as its cardinality.

2. Problem Formulation

We model the emergency department (ED) as a multi-server queue with a potentially time-varying number of servers, denoted by $c(t)$. Patients arrive through two distinct streams: (1) External arrivals are patients entering the ED from outside the hospital system, such as walk-ins, ambulance arrivals (not from another ED), and referrals from external healthcare providers; and (2) Transfer requests are patients seeking transfer from other EDs. External arrivals follow a time-varying Poisson process with rate $\lambda_1(t)$, while transfer requests arrive according to a Poisson process with rate $\lambda_2(t)$. We denote the total arrival rate by $\lambda(t) = \lambda_1(t) + \lambda_2(t)$. Patients are served on a first-come-first-served basis, with service times that are exponentially distributed with rate μ . While waiting for service, patients may abandon the queue if their patience expires; patience times are assumed to be exponentially distributed with rate $\theta < \mu$. Upon completing ED treatment, a patient either leaves the hospital directly with probability p or is admitted to an inpatient ward (IW) with probability $1 - p$. Patients requiring inpatient admission experience a boarding delay prior to transfer, reflecting limited bed availability or administrative bottlenecks on the inpatient side. These boarding times are modeled as exponentially distributed with a time-varying rate $\nu(t)$, which we refer to as the processing rate. Importantly, patients remain in an ED service position while boarding and therefore continue to occupy ED server capacity until their transfer to the inpatient ward is completed.

We consider an admission control policy under which the hospital determines whether to accept incoming transfer requests. Specifically, the control variable $G(t) \in \{0, 1\}$ indicates whether transfer requests arriving at time t are accepted, $G(t) = 1$, or rejected, $G(t) = 0$.

Let $X(t)$ denote the total number of patients in the ED who have not yet completed service (treatment), including both those waiting in the queue and those currently receiving service. In addition, let $Z(t)$ denote the number of patients currently in service and $Q(t)$ denote the number of patients waiting in the queue. Then, $X(t) = Q(t) + Z(t)$. We further let $B(t)$ denote the number of patients boarding in the ED while awaiting transfer to the IWs.

Let A_1 , A_2 , S , D , and R be independent rate-1 Poisson processes representing external arrivals, transfer requests, service completions, patient abandonments, and inpatient transfer completions, respectively. The system dynamics can then be characterized by

$$\begin{aligned} X(t) &= X(0) + A_1 \left(\int_0^t \lambda_1(t) dt \right) + A_2 \left(\int_0^t \lambda_2(t) G(t) dt \right) - S \left(\int_0^t \mu Z(t) dt \right) - D \left(\int_0^t \theta Q(t) dt \right), \\ B(t) &= B(0) + S \left(\int_0^t (1-p) \mu Z(t) dt \right) - R \left(\int_0^t \nu(t) B(t) dt \right), \end{aligned} \quad (1)$$

Without loss of optimality, we assume the system is work-conserving, so that

$$Z(t) = \min\{c(t) - B(t), X(t)\}.$$

An admission control policy $G(t)$ may depend on the current system state $(X(t), B(t))$ and on information available at time t about the future operating environment. In particular, let $\hat{\lambda}_i(s | t)$, $i \in 1, 2$, and $\hat{\nu}(s | t)$, $s \geq t$, denote forecasts of future arrival and inpatient processing rates constructed using information available at time t . We assume these forecasts are available to support admission decisions. We also assume that future staffing levels $\{c(s), s \geq t\}$ are known, reflecting the fact that staffing schedules are typically determined in advance. Importantly, these forecasts describe the anticipated evolution of the system primitives; they do not reveal future sample-path realizations of individual arrivals, service completions, or inpatient transfers.

Let π denote an admissible admission control policy, and use superscripts to indicate dependence on π , e.g., $X^\pi(t)$, $B^\pi(t)$. We associate a holding cost h per patient per unit time spent waiting in the queue, and a blocking cost l for each rejected transfer request. The objective is to minimize the expected total cost over a planning horizon T :

$$V^\pi(x, b) = \mathbb{E} \left[\int_0^T h(X^\pi(t) - Z^\pi(t)) dt + lL^\pi(T) \mid X(0) = x, B(0) = b \right] \quad (2)$$

where $L^\pi(T)$ denotes the number of transfer requests rejected by time T . Note that

$$L^\pi(T) \sim \tilde{E} \left(\int_0^T \lambda_2(t) (1 - G^\pi(t)) dt \right) \quad (3)$$

where $\tilde{E}$ is a rate-1 Poisson process. The planning horizon T is chosen to be sufficiently long for the system to fully recover from any demand surge; a formal definition is provided in Appendix B.

The model deliberately abstracts from several operational details of ED patient flow, such as unit-level bed heterogeneity, patient-level acuity, and service prioritization. Our objective is not to reproduce the full complexity of ED operations but to isolate the central trade-off in transfer admission control: accepting a transfer increases future congestion, whereas rejecting it incurs an immediate blocking cost. This trade-off is primarily driven by the interaction among the current ED and boarding census, anticipated future demand, and expected inpatient processing conditions. This level of aggregation is also consistent with practice, where transfer-admission decisions are typically based on system-level indicators such as ED census, boarding levels, and expected discharges rather than detailed patient- or unit-level information. The model's state variables therefore align with both the information routinely available to decision makers and the organizational level at which transfer-admission decisions are implemented.

3. Fluid Analysis

For analytical development, we first consider a deterministic reference model in which the future primitive trajectories $\{\lambda_1(s), \lambda_2(s), \nu(s), c(s) \geq t\}$ are treated as given. This reference model allows us to characterize how anticipated future operating conditions should enter the admission decision. Even under this reference setting, the stochastic control problem in (2) remains analytically intractable because of (i) the nonstationary primitives $(\lambda_1(t), \lambda_2(t), \nu(t), c(t))$ and (ii) the complexity induced by the joint evolution of the ED census and boarding population. Moreover, even when numerical solutions can be obtained for stylized instances, they provide limited structural insight into how forecasted operating conditions should influence admission decisions.

To obtain a tractable characterization of the optimal policy, we therefore study a deterministic fluid approximation. The fluid model captures the mean dynamics of the underlying queueing system and is particularly well suited to the congested, time-varying regimes of interest. Importantly, the fluid analysis allows us to identify a low-dimensional summary of future operating conditions that governs the admission decision, which will later serve as the basis for the forecast-based policy implemented in the stochastic system.

Let $x(t)$, $z(t)$, $b(t)$, and $g(t)$ denote the fluid analogues of $X(t)$, $Z(t)$, $B(t)$, and $G(t)$, respectively. The fluid dynamics satisfy

$$\begin{aligned}\dot{x}(t) &= \lambda_1(t) + g(t)\lambda_2(t) - \mu z(t) - \theta(x(t) - z(t)), \\ \dot{b}(t) &= (1-p)\mu z(t) - \nu(t)b(t),\end{aligned}\tag{4}$$

where $z(t) = \min\{c(t) - b(t), x(t)\}$ and $\dot{x}(t) = dx(t)/dt$, $\dot{b}(t) = db(t)/dt$.

3.1. Single Congestion Shock

We first analyze a setting with a single congestion shock beginning at time 0 and ending at time $\kappa > 0$. Such a shock may be driven by a demand surge, a temporary reduction in the processing rate, or both. We impose the following regularity and load conditions.

ASSUMPTION 1. 1. The functions $\lambda_1(t)$, $\lambda_2(t)$, $\nu(t)$, and $c(t)$ are piecewise monotone with a finite number of pieces.

2. The initial condition satisfies $x(0) + b(0) \geq c(0)$. There exists $\epsilon_1 > 0$ such that for all $t \in [0, \kappa)$,

$$\lambda_1(t) - \max\{p\mu c(t), \nu(t)c(t)\} \geq \epsilon_1.$$

3. There exists $\epsilon_2 > 0$ such that for all $t \geq \kappa$,

$$\lambda(t) - \min\{p\mu c(t), \nu(t)c(t)\} \leq -\epsilon_2,$$

where $\lambda(t) = \lambda_1(t) + \lambda_2(t)$.

Assumption 1.1 is a standard regularity condition ensuring existence and uniqueness of solutions to the fluid ODEs and enabling convergence arguments under many-server scaling. Assumptions 1.2–3 impose a pronounced overload prior to time κ and a subcritical regime thereafter, which together ensure that the fluid queue remains positive before κ and is eventually cleared after κ . While these conditions can be relaxed, the essential requirement for our analysis is that the fluid queue does not hit zero during the overload phase and eventually drains to zero in finite time.

Consistent with the deterministic reference model, we assume when choosing $g(t)$, the decision maker treats the future primitive trajectories $\{\lambda_1(s), \lambda_2(s), \nu(s), c(s) : s \geq t\}$ as given.

Let $q(t) = x(t) - z(t)$ denote the fluid queue length. Define the (fluid) queue-clearing time

$$\tau = \inf\{t \geq \kappa : x(t) + b(t) = c(t)\}.$$

Under Assumption 1, $\tau < \infty$ and $q(t) = 0$ for all $t \geq \tau$. The fluid optimal control problem corresponding to (2) can be written as

$$\begin{aligned} \min_{g(\cdot)} \quad & \int_0^\tau \left[h(x(t) - z(t)) + l\lambda_2(t)(1 - g(t)) \right] dt \\ \text{s.t.} \quad & (4), \quad z(t) = \min\{x(t), c(t) - b(t)\}, \quad 0 \leq g(t) \leq 1. \end{aligned} \quad (5)$$

For $t \geq 0$, define the hitting time $H_t(x, b)$ as the time required for the system to return to a zero-queue state when *all* transfers are accepted (i.e., $g(\cdot) \equiv 1$), starting from state (x, b) at time t :

$$H_t(x, b) := \inf \left\{ \Delta > 0 : \tilde{x}(t + \Delta) + \tilde{b}(t + \Delta) \leq c(t + \Delta) \mid \tilde{x}(t) = x, \tilde{b}(t) = b \right\}, \quad (6)$$

where $(\tilde{x}(\cdot), \tilde{b}(\cdot))$ is the solution to the uncontrolled ODE system

$$\begin{aligned} \tilde{x}(t) &= x, \quad \tilde{b}(t) = b, \\ \dot{\tilde{x}}(u) &= \lambda(u) - \mu(c(u) - \tilde{b}(u)) - \theta \tilde{q}(u), \quad u \geq t, \\ \dot{\tilde{b}}(u) &= (1 - p)\mu(c(u) - \tilde{b}(u)) - \nu(u)\tilde{b}(u), \quad u \geq t, \end{aligned}$$

with $\tilde{q}(u) = \tilde{x}(u) - \min\{\tilde{x}(u), c(u) - \tilde{b}(u)\}$. For boundary states satisfying $x + b = c(t)$, we define $H_t(x, b)$ by continuity as $H_t(x, b) = \lim_{x \downarrow (c(t) - b)} H_t(x, b)$. Intuitively, $H_t(x, b)$ measures the duration of congestion implied by the current state and the future primitives $(\lambda(\cdot), \nu(\cdot), c(\cdot))$.

The following theorem characterizes the optimal admission control for (5).

THEOREM 1. *Under Assumption 1, an optimal control for (5) is given by*

$$g^*(t) = \mathbf{1} \left\{ \frac{h}{\theta} \left(1 - \exp(-\theta H_t(x(t), b(t))) \right) \leq l \right\}. \quad (7)$$

The optimal decision can be understood by comparing the two costs associated with an arriving transfer patient. Rejecting the patient incurs the immediate blocking cost l . Accepting the patient, by contrast, creates a marginal congestion cost: while the ED remains congested, the additional patient contributes to the queue and generates waiting cost until either the patient abandons or the congestion episode ends. The magnitude of this cost therefore increases with the remaining congestion horizon. When recovery is far away, the additional patient is expected to contribute to congestion for longer; as the system approaches clearing, this marginal burden declines. This monotonic decline in the marginal congestion cost gives rise to a single switching structure. Early in the congestion episode, rejection is optimal when the marginal congestion cost exceeds the blocking cost. As recovery approaches, the marginal congestion cost falls, and the policy eventually switches to acceptance. Once this switch occurs, all subsequent transfers are accepted, so the actual remaining congestion duration coincides with the all-accept hitting time H_t . The optimal policy can therefore be expressed directly as a threshold in H_t .

The corresponding congestion cost has a simple interpretation. With exponentially distributed patience and abandonment rate θ , the probability that the marginal patient remains waiting s time units later is $e^{-\theta s}$. Hence, over a congestion horizon H_t , the expected waiting time contributed by this patient before either abandonment or queue clearing is

$$\int_0^{H_t} e^{-\theta s} ds = \frac{1 - e^{-\theta H_t}}{\theta}.$$

Multiplying by the holding-cost rate h yields the marginal congestion cost. This expression also clarifies two limiting cases. If $h/\theta \leq l$, then even the maximum possible expected waiting cost of an additional patient—obtained when congestion persists indefinitely—is no greater than the blocking cost. Hence, it is never optimal to reject a transfer patient. In the special case of no abandonment, $\theta = 0$, taking the limit $\theta \downarrow 0$ yields the simpler threshold $hH_t(x(t), b(t)) \leq l$. More broadly, the relevant cost of admitting an additional transfer is determined not by the instantaneous queue length alone, but by the expected duration of congestion implied by the current state and anticipated future operating conditions.

The detailed proof of Theorem 1 is provided in Appendix A.1, where we apply Pontryagin's Minimum Principle and verify the sufficient conditions for optimality.

3.2. Multiple Congestion Shocks

The single-shock setting in Section 3.1 yields a particularly transparent characterization because the system experiences only one congestion episode before fully clearing. In practice, however, the ED may encounter multiple congestion episodes over the planning horizon due to repeated

demand surges or recurrent reductions in inpatient processing rate. In this subsection, we extend the analysis to this more general setting. Our goal is twofold. First, we show that the optimal admission policy in the multi-surge setting still has a threshold structure. Second, we explain why the exact threshold statistic is substantially more complicated than in the single-surge case, and motivate a simpler approximation based on the single-surge policy.

To formulate the multi-surge setting, we impose a piecewise-overload structure analogous to Assumption 1.

ASSUMPTION 2 (Piecewise overload/underload structure). *The functions $\lambda_1(t)$, $\lambda_2(t)$, $\nu(t)$, and $c(t)$ are piecewise monotone with a finite number of pieces. Moreover, there exist times*

$$0 = \kappa_0 < \kappa_1 < \dots < \kappa_M < \infty$$

such that on each interval $[\kappa_{m-1}, \kappa_m)$, the net input condition is of a fixed sign; that is, the system is either overloaded or underloaded throughout the interval, i.e., either

$$\lambda_1(t) - \max\{p\mu c(t), \nu(t)c(t)\} \geq \epsilon_m^+$$

for some $\epsilon_m^+ > 0$, or

$$\lambda(t) - \min\{p\mu c(t), \nu(t)c(t)\} \leq -\epsilon_m^-$$

for some $\epsilon_m^- > 0$, where $\lambda(t) := \lambda_1(t) + \lambda_2(t)$.

A key distinction from the single-surge case is that the timing of congestion is endogenous. Even if the instantaneous load becomes unfavorable at some time, the queue may take time to build up before becoming positive; similarly, once the load turns favorable again, the queue may persist for some time before it fully dissipates. It is therefore useful to distinguish between primitive stress intervals, defined from the model inputs, and the resulting *congestion periods*, defined from the fluid state. Recall that the fluid queue length is

$$q(t) = x(t) - z(t) = (x(t) + b(t) - c(t))^+.$$

Let $(x(\cdot), b(\cdot))$ be the fluid trajectory induced by an admissible control $g(\cdot)$. A *congestion period* is a maximal interval $I_k = [\sigma_k, \tau_k]$ such that $q(t) > 0$ for $t \in (\sigma_k, \tau_k)$. An *uncongestion period* is a maximal interval $G_k = [\tau_k, \sigma_{k+1}]$ such that $q(t) = 0$ for $t \in G_k$. Thus, a congestion period is an interval during which a positive queue is present, whereas an uncongestion period is an interval during which the queue has fully cleared.

3.2.1. Two-Shock Case We first consider the simplest multi-shock environment in which there are two congestion shocks. Let $I_1 = [0, \tau_1]$, i.e., $\sigma_1 = 0$, and $I_2 = [\sigma_2, \tau_2]$ denote the two congestion periods, separated by the uncongested gap $G_1 = [\tau_1, \sigma_2]$. Note that if no such uncongested gap exists, then the two shocks effectively merge into a single congestion episode, reducing the problem to the single-shock case.

Let $p_x(t)$ and $p_b(t)$ denote the adjoint processes associated with the state variables $x(t)$ and $b(t)$ along an optimal trajectory, as given by Pontryagin's Maximum Principle; see Appendix A for the full adjoint system and transversality conditions. These adjoint variables encode the marginal value of perturbations to the fluid state: $p_x(t)$ is the marginal effect on the objective of an additional ED arrival at time t , while $p_b(t)$ is the marginal effect of an additional boarding patient at time t . As in the single-shock case, the Hamiltonian is affine in the control variable $g(t)$; see Appendix A. Therefore, the optimal control has threshold form: $g^*(t) = \mathbf{1}\{p_x(t) \leq l\}$. It is convenient to define the *effective congestion horizon*

$$H_t^e := f^{-1}(p_x(t)), \text{ where } f(H) := \frac{h}{\theta}(1 - e^{-\theta H}).$$

Then, the optimal policy can be written equivalently as $g^*(t) = \mathbf{1}\{f(H_t^e) \leq l\}$.

THEOREM 2 (Optimal policy in the two-surge case). *In the two-surge setting described above, the optimal admission policy is given by*

$$g^*(t) = \mathbf{1}\{p_x(t) \leq l\} = \mathbf{1}\{f(H_t^e) \leq l\}, \quad t \geq 0.$$

The effective congestion horizon H_t^e admits the following characterization:

(i) **Congestion periods.** For $t \in I_k = [\sigma_k, \tau_k]$, $k = 1, 2$,

$$H_t^e = (\tau_k - t) + C_k, \quad C_k := H_{\tau_k}^e.$$

In particular, $C_2 = 0$, while C_1 captures the impact of the second congestion episode on the first.

(ii) **Uncongested period.** For $t \in G_1 = [\tau_1, \sigma_2]$, the adjoint variables satisfy

$$p_b(t) = \exp\left(-\int_t^{\sigma_2} \nu(r) dr\right) p_b(\sigma_2),$$

and

$$p_x(t) = e^{-\mu(\sigma_2 - t)} p_x(\sigma_2) + (1 - p) \mu \int_t^{\sigma_2} e^{-\mu(u - t)} \exp\left(-\int_u^{\sigma_2} \nu(r) dr\right) du p_b(\sigma_2).$$

Theorem 2 shows that future congestion episodes influence current admission decisions through carry-over terms propagated backward across time. This structure extends naturally to a general multi-shock setting. In particular, for $k = 1, \dots, K$, on a congestion period $I_k = [\sigma_k, \tau_k]$,

$$H_t^e = (\tau_k - t) + C_k, \text{ for } t \in I_k,$$

where $C_k = H_{\tau_k}^e$ summarizes the effect of subsequent congestion episodes. Thus, the effective congestion horizon consists of the remaining duration of the current congestion episode plus a continuation term capturing future congestion. During an uncongestion period G_k , the queue is empty, but an additional patient present at time $t \in G_k$ may still affect the system when the next congestion episode begins. Let $p_x(\sigma_{k+1})$ and $p_b(\sigma_{k+1})$ denote the backward terminal values at the onset of the next congestion period. Then, for $t \in G_k$,

$$p_b(t) = \exp\left(-\int_t^{\sigma_{k+1}} \nu(r) dr\right) p_b(\sigma_{k+1}),$$

and

$$p_x(t) = e^{-\mu(\sigma_{k+1}-t)} p_x(\sigma_{k+1}) + (1-p)\mu \int_t^{\sigma_{k+1}} e^{-\mu(u-t)} \exp\left(-\int_u^{\sigma_{k+1}} \nu(r) dr\right) du p_b(\sigma_{k+1}). \quad (8)$$

The two terms in (8) capture the two channels through which a patient present at time t can carry congestion effects into the next episode. The first corresponds to the patient remaining in ED treatment until σ_{k+1} . The second corresponds to the patient completing ED treatment at some $u \in [t, \sigma_{k+1}]$, requiring inpatient admission, and remaining boarded in the ED until σ_{k+1} .

Representation (8) also shows that this intertemporal carry-over weakens as the uncongested gap becomes longer. In particular, if $\nu(r) \geq \underline{\nu} > 0$ on G_k , both propagation channels are exponentially attenuated with the distance to the next congestion episode. Consequently, the carry-over term C_k becomes small as $\sigma_{k+1} - \tau_k$ increases. Intuitively, a long uncongested interval gives the system time to absorb the effect of current admissions before the next surge arrives, so sufficiently distant future congestion has only a limited influence on current admission decisions.

3.2.2. Accuracy of the H_t -Based Policy The exact effective horizon H_t^e incorporates the influence of all future congestion episodes and therefore requires solving the adjoint system backward across each congested and uncongested phase. This dependence on the full future switching structure makes H_t^e difficult to compute in real time. For operational use, we therefore consider the simpler H_t -based policy $g^H(t) = \mathbf{1}\{f(H_t) \leq l\}$, where H_t is the all-accept queue-clearing time defined in (6). Unlike H_t^e , which incorporates the carry-over effect of later congestion episodes, H_t summarizes only the congestion implied by the current state and the local all-accept recovery trajectory. We next assess the quality of this approximation in the two-surge setting by comparing g^H with the exact optimal control g^* .

Let $J(g)$ denote the fluid objective under an admissible control $g(\cdot)$. Directly characterizing the exact gap $J(g^H) - J(g^*)$ is difficult because changing the control alters both the state trajectory and the future switching structure of the optimal-control problem. We therefore study the first-order loss obtained by perturbing the optimal control in the direction $g^H - g^*$.

Let $(x^*(t), b^*(t))$ denote the state trajectory under g^* , and recall that $p_x(t)$ is the corresponding adjoint associated with the state variable $x(t)$ along this optimal trajectory. Define

$$\Delta J^{(1)}(g^H; g^*) := \int_0^{\tau_2} \lambda_2(t) (p_x(t) - l) (g^H(t) - g^*(t)) dt. \quad (9)$$

Lemma 1 below shows that this is precisely the directional derivative of J at g^* in the direction $g^H - g^*$. The integrand in (9) has a natural interpretation. The term $p_x(t) - l$ is the marginal net cost of accepting rather than rejecting a transfer request at time t : $p_x(t)$ is the marginal future congestion cost generated by one additional ED patient, while l is the blocking cost of rejecting that transfer. The term $g^H(t) - g^*(t)$ identifies times at which the proxy and optimal rules disagree, and $\lambda_2(t)$ weights these mismatches by the rate at which transfer requests arrive. Hence, $\Delta J^{(1)}(g^H; g^*)$ aggregates the first-order cost of all decision mismatches over the horizon.

LEMMA 1 (**First-variation representation**). *For every admissible control g , the directional derivative of the fluid objective J at g^* in the direction $g - g^*$ exists and is given by*

$$DJ(g^*; g - g^*) = \int_0^{\tau_2} \lambda_2(t)(p_x(t) - l)(g(t) - g^*(t)) dt.$$

Lemma 1 implies $\Delta J^{(1)}(g^H; g^*) = DJ(g^*; g^H - g^*)$, so $J(g^H) - J(g^*) = \Delta J^{(1)}(g^H; g^*) +$ higher-order terms.

Let $K := f^{-1}(l)$ denote the critical congestion horizon at which the marginal congestion cost equals the rejection cost. In the single-surge setting, transfers are accepted if and only if the remaining congestion horizon does not exceed K . Recall also that $C_1 = f^{-1}(p_x(\tau_1))$, which captures the carry-over effect of the second congestion episode on decisions during the first. We assume $C_1 \leq K$, so that the carry-over effect shifts the switching boundary by at most K .

PROPOSITION 1 (**First-order loss of the H_t -based rule**). *Assume that $\mu \neq \nu$, $\lambda_2(t) \equiv \lambda_2$ on $[0, \tau_2]$, $\nu(t) \equiv \nu$ on $[\tau_1, \tau_2]$, and $C_1 \leq K$. Let $t_A := \sup\{t \in [\tau_1, \sigma_2] : p_x(t) \leq l\}$, with the convention $t_A = \tau_1$ if the set is empty. Then, the first-order proxy of g^H relative to g^* decomposes as*

$$\Delta J^{(1)}(g^H; g^*) = \Delta J^{\text{gap}} + \Delta J^{\text{carry}},$$

where

$$\Delta J^{\text{gap}} = \lambda_2 \int_{t_A}^{\sigma_2} (p_x(t) - l) dt, \quad \Delta J^{\text{carry}} = \lambda_2 \int_{K-C_1}^K (f(u+C_1) - f(K)) du.$$

Moreover, if $\tau_2 - \sigma_2 \leq K$, then $\Delta J^{\text{gap}} = 0$, and, as $C_1 \downarrow 0$,

$$\Delta J^{\text{carry}} = \frac{\lambda_2 h}{2} e^{-\theta K} C_1^2 + O(C_1^3).$$

Proposition 1 identifies two distinct sources of approximation error. The first, ΔJ^{gap} , arises during the uncongested gap: because $H_t = 0$ once the current queue has cleared, the H_t -based rule does not proactively reject transfers in anticipation of the second congestion episode, whereas the optimal rule may do so. The second, ΔJ^{carry} , arises during the first congestion period: H_t omits the continuation effect of the second episode and therefore understates the effective congestion horizon by the carry-over term C_1 . The proposition also clarifies when these omissions are innocuous. The gap term vanishes when the second congestion episode is too short to make proactive rejection worthwhile. The carry-over term is second order in C_1 , and C_1 itself becomes small when the intervening uncongested period is long. Thus, the H_t -based rule is expected to be accurate when the next congestion episode is either mild or sufficiently distant.

To assess whether these first-order insights translate into meaningful performance differences in the stochastic system, we further compare the H_t -based rule with the fluid-optimal policy numerically. Both policies are implemented in the two-surge stochastic system described in Section 5, with review interval $\delta = 1$. Figure 1 compares the performance of the two policies. The left panel traces the efficiency frontier between average waiting time and the fraction of rejected transfer

requests as l/h varies. The H_t -based frontier closely tracks the fluid-optimal frontier, indicating little loss from replacing H_t^e with the simpler local hitting-time statistic H_t . The right panel reports total cost, with $h = 1$, together with 95% confidence bands based on 200 independent simulation replications. The two policies yield nearly indistinguishable costs over most of the range of l , with a small gap only when rejection is relatively inexpensive and the admission decision is therefore more sensitive to omitted future-congestion effects. Overall, the H_t -based rule captures nearly all of the performance of the fluid-optimal control while avoiding the need to solve the full backward adjoint system.

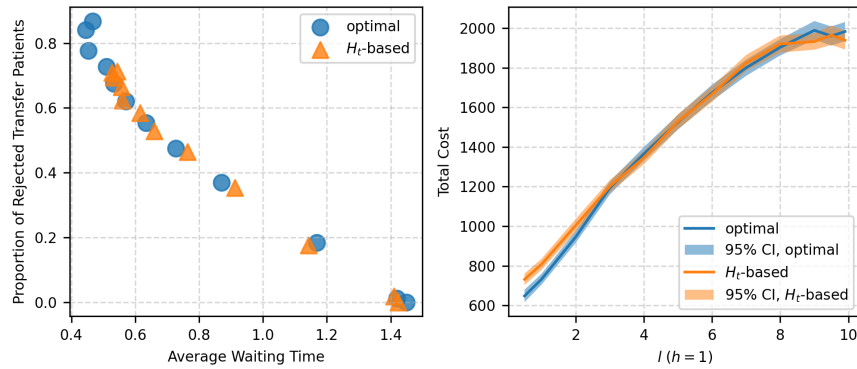


Figure 1 Comparison between H_t -based policy (orange) and fluid-optimal policy (blue).

3.3. Extensions to Patient Heterogeneity

The structural insights of Section 3.1 extend to certain forms of patient heterogeneity. Suppose transfer classes differ only in their rejection costs l_k , while sharing the same service, abandonment, inpatient-admission, and holding-cost parameters. An admitted patient then generates the same marginal congestion cost regardless of class, and the optimal decision compares this common cost with the class-specific rejection cost l_k . Similarly, if patients differ only in ED service priority, work conservation leaves the aggregate fluid dynamics unchanged. Priority determines which patients receive service but does not change the aggregate congestion cost.

Section 6.1 considers an extension motivated by the empirical setting: external and transfer patients have different probabilities of requiring inpatient admission, and transfers receive priority for ED service. These differences affect both the fluid dynamics and the marginal congestion cost of admitting a transfer. We use the heterogeneous fluid model to predict the time until congestion clears and apply the hitting-time rule from Section 3.1 as a heuristic.

Appendix E allows for broader differences between patients. It considers high- and low-priority classes, denoted by H and L , with preemptive priority for class H . External arrivals and transfer requests may have different class compositions, and transfer admission decisions are made separately by class. The classes may differ in service and abandonment rates, probabilities of requiring

inpatient admission, and holding and rejection costs. Admitting a high-priority patient can delay lower-priority patients, so a single congestion-clearing time cannot capture every admission consequence. We therefore develop a class-specific look-ahead heuristic that accounts for these cross-class effects: it accepts a transfer when the predicted congestion cost of admission does not exceed the cost of rejection.

4. Admission Control Policies for the Stochastic System

In this section, we describe how the fluid-optimal policy can be implemented in the original stochastic system and discuss several practical considerations that arise in operational settings, including prediction error and limited decision-update frequency.

4.1. Fluid-Inspired Admission Control Rule

The fluid analysis yields a simple, interpretable admission rule that can be directly applied to the stochastic system. At any time t , the decision maker evaluates the condition

$$\frac{h}{\theta} \left(1 - \exp \left(-\theta H_t(X(t), B(t)) \right) \right) > l.$$

If the inequality holds, incoming transfer requests are rejected; otherwise, they are accepted. Here, $(X(t), B(t))$ denote the current stochastic system state, and $H_t(\cdot)$ is the congestion hitting-time metric computed from the fluid model. Operationally, $H_t(X(t), B(t))$ represents the predicted time required for congestion to clear if all transfer requests were accepted from time t onward.

This policy has several appealing features from an implementation standpoint. First, it depends only on low-dimensional, routinely monitored state variables (ED census and boarding population). Second, future arrival rates, processing rates, and staffing levels enter the decision only through the scalar quantity H_t , providing a natural way to aggregate predictive information into a single operational signal.

Although the admission control rule is derived from a deterministic fluid approximation, it admits a rigorous justification for the underlying stochastic system. In particular, under a many-server scaling regime in which arrival rates and capacity grow proportionally, the fluid-translated policy is asymptotically optimal. This result provides formal theoretical support for using the fluid hitting time as the decision-relevant quantity for admission control in large, highly-utilized EDs. The precise statement of the asymptotic optimality result, along with the technical assumptions and proof, is provided in Appendix B.

4.2. Prediction Error and Imperfect Information

In practice, future arrival and inpatient processing rates are not known exactly and must be forecast using information available at the time of decision. Let $\hat{\lambda}(s | t)$ and $\hat{\nu}(s | t)$, $s \geq t$, denote the forecasts available at review epoch t . We implement the fluid-based policy in a certainty-equivalent

manner: conditional on these forecast trajectories, we evaluate the deterministic fluid dynamics and compute the corresponding predicted congestion-clearing time, denoted by $\hat{H}_t$. The admission rule then rejects transfer requests whenever

$$\frac{h}{\theta} \left(1 - \exp \left(-\theta \hat{H}_t(X(t), B(t)) \right) \right) > l.$$

The resulting decision is implemented until the next review epoch, at which point the system state and forecasts are updated and the hitting time is recomputed. Thus, the policy requires forecasts only of aggregate future arrival and processing rates. Accordingly, the fluid optimality result should be interpreted as conditional on a given trajectory of the system primitives, while Sections 5 and 6 examine the performance implications of replacing these trajectories with imperfect and data-driven forecasts.

When detailed forecasts of the future rate trajectories are unavailable, the same framework can be implemented using coarser estimates of the anticipated operating environment. Let $\bar{\lambda}_t$ and $\bar{\nu}_t$ denote baseline estimates of the future arrival and inpatient processing rates constructed using information available at time t , for example from historical averages for comparable time periods or other simple forecasting models. We then compute an approximate hitting time $\bar{H}_t$ by evaluating the fluid dynamics under constant future rates $\hat{\lambda}(s | t) = \bar{\lambda}_t$ and $\nu(s | t) = \bar{\nu}_t$ for $s \geq t$. This provides a simple fallback implementation when detailed time-varying forecasts are unavailable.

Several features of the policy help mitigate the impact of prediction error. First, the current system state $(X(t), B(t))$ provides real-time information about congestion, reducing the policy's reliance on forecasts alone. Second, the hitting-time metric summarizes the predicted arrival and processing rates over the future congestion horizon, so small, transitory forecast errors may have only a limited effect on the resulting estimate. Third, the admission decision depends only on whether the predicted hitting time crosses a fixed threshold. Consequently, moderate forecast errors are unlikely to change the decision when the system is sufficiently far from the switching boundary. When the predicted hitting time is close to the threshold, forecast errors may change the admission decision; however, in this region the marginal congestion cost of accepting a transfer is close to the blocking cost of rejecting it, so the associated performance loss is expected to be small. These observations suggest that the policy should be relatively robust to modest forecast errors, while persistent biases or substantial errors in the magnitude or timing of future congestion may lead to more meaningful performance degradation. We examine these effects numerically in Section 5.2.

4.3. Periodic Review and Implementation Considerations

In operational environments, admission decisions are rarely updated continuously. Moreover, frequent re-evaluation can lead to undesirable decision volatility driven by stochastic fluctuations in the system state, causing the hitting-time metric H_t to oscillate around the decision threshold.

To address these practical considerations, we consider a periodic review implementation in which admission decisions are updated at discrete intervals of length δ .

Specifically, at review times $k\delta$, $k = 0, 1, 2, \dots$, the admission decision is fixed over the interval $[k\delta, (k+1)\delta)$. Transfer requests are rejected during this interval if

$$H_t(X(k\delta), B(k\delta)) - a\delta > -\frac{1}{\theta} \log \left(1 - \frac{l\theta}{h} \right), \quad (10)$$

where $a \geq 0$ is a tuning parameter introduced to compensate for the time lag between reviews.

The review interval δ governs a trade-off between responsiveness and stability. Smaller values of δ allow the policy to react quickly to changes in congestion, but may result in frequent switching driven by noise. Larger values of δ yield more stable decisions and lower administrative burden. Importantly, because the policy depends on whether H_t exceeds a fixed threshold, decision quality is relatively insensitive to δ when H_t is well above or well below the threshold. When H_t is close to the threshold, the performance difference between accepting and rejecting transfers is small, so less frequent updates can help reduce volatility without materially affecting outcomes. We illustrate these effects through numerical experiments in Sections 5 and 6.

5. Numerical Experiments

In this section, we evaluate the performance of the proposed look-ahead admission control policy through a set of synthetic numerical experiments. Our primary objective is to quantify the operational value of incorporating predictive information into transfer admission decisions, relative to policies that rely solely on contemporaneous congestion measures. Because admission control inherently involves a trade-off between congestion mitigation and access for transfer patients, policy performance cannot be meaningfully evaluated at a single operating point. Accordingly, throughout this section we summarize performance using trade-off curves between the average waiting time and the fraction of rejected transfer requests.

Policy comparison. We compare the proposed look-ahead policy against a class of state-dependent benchmark policies motivated by current hospital practice and by optimal admission control rules in queueing models without predictive information. These benchmark policies rely exclusively on contemporaneous congestion indicators and do not incorporate forecasts of future demand or processing capacity.

Under a benchmark policy, an incoming transfer request is rejected whenever $\eta X(t) + B(t) > \Gamma$, where $X(t)$ denotes the total ED census, $B(t)$ the boarding population, and η and Γ are tunable parameters. This class captures a wide range of commonly used threshold rules based on ED occupancy and boarding levels.

For any fixed η , varying Γ traces out the achievable trade-off between congestion (measured by the average waiting time) and access (measured by transfer rejection rate) within this class

of state-based policies. The resulting trade-off curve summarizes the attainable performance of policies that rely only on contemporaneous system information and do not use predictive inputs, providing a natural and fair benchmark for assessing the incremental value of prediction.

We evaluate benchmark policies for $\eta \in \{0, 0.5, 1\}$ and $\Gamma \in \{10, 11, \dots, 100\}$. Once Γ is appropriately tuned, benchmark performance is largely insensitive to the choice of η (see Appendix D for details). For clarity, we therefore present results for $\eta = 0$.

For the proposed look-ahead policy, the numerical computation details for H_t are provided in Appendix D, and we construct an analogous trade-off curve by varying the ratio l/h , which governs the relative penalty placed on transfer rejection versus patient waiting.

Experimental design and baseline parameters. Unless otherwise noted, we consider a system with constant service capacity $c(t) = 30$ over a horizon $T = 150$. A fraction $\gamma = 0.2$ of arrivals correspond to transfer requests, with the remaining arrivals representing non-transfer patients. Service times are exponential, with the service rate normalized to $\mu = 1$, so that time is measured in units of mean service time. The abandonment rate is $\theta = 0.1$, and patients are discharged directly from the ED with probability $p = 0.5$. The system is initialized at $(X(0), B(0)) = (30, 0)$. We normalize the waiting cost to $h = 1$ and vary the blocking cost $l \in [0, 10)$ to trace out the tradeoff curves. Admission decisions are updated periodically, with review intervals $\delta \in \{1, 5\}$ and $a = 0.5$ to compensate for the time lag between reviews. All reported performance measures are estimated as averages over 200 independent sample paths.

In the main paper, we fix the inpatient processing rate at $\nu(t) = 0.5$ to isolate the effect of predictive demand information. Appendix D considers scenarios with time-varying and reduced processing rates; across these settings, the qualitative comparison between policies remains unchanged. We assume the processing rate is known without error, i.e., $\hat{\nu}(s | t) = \nu(s)$.

Arrival pattern and prediction error. We assess policy performance along two dimensions. First, we consider arrival patterns corresponding to a single congestion surge and two congestion surges, which capture distinct temporal structures of demand shocks. Second, and more importantly, we examine how different forms of prediction error in arrival rate forecasts affect policy performance.

In all congestion-surge scenarios, admission decisions are based on predicted arrival rates rather than perfect future information. Predicted arrival rates are generated on a discrete time grid with step size $\Delta t = 0.1$ and held constant between updates, with nonnegativity enforced by truncation.

5.1. Congestion surge scenarios

To isolate the value of forecast information from the details of any particular prediction method, we begin with a stylized baseline forecast-error model. At each review epoch, the policy is supplied with a forecast trajectory of future arrival rates obtained by perturbing the underlying rate trajectory

with unbiased noise. This provides a controlled setting for evaluating the policy under imperfect forecasts before considering richer forms of forecast misspecification in Section 5.2.

We examine two demand environments that differ in the temporal structure of congestion: a single-surge scenario and a two-surge scenario. In both cases, the baseline forecast errors are assumed to be unbiased and independently distributed. For each review epoch t , let $s_k = t + k\delta t$, $k = 0, 1, \dots$, denote the future forecast grid,

$$\hat{\lambda}(s_k | t) = \lambda(s_k) + \epsilon_{t,k}, \quad \epsilon_{t,k} \sim \mathcal{N}(0, \sigma^2).$$

Thus, the forecasts preserve the underlying temporal pattern of demand while introducing noise in the predicted arrival rates.

5.1.1. Single congestion shock We first consider a scenario with a single congestion shock, where the arrival rate $\lambda(t)$ evolves as

$$\lambda(t) = \begin{cases} 2, & 0 \leq t \leq 50, \\ 20, & 50 < t \leq 100, \\ 2, & 100 < t \leq T, \\ 0, & \text{otherwise.} \end{cases}$$

The left panel of Figure 2 shows a realization of the predicted arrival rate $\hat{\lambda}(t | 0)$.

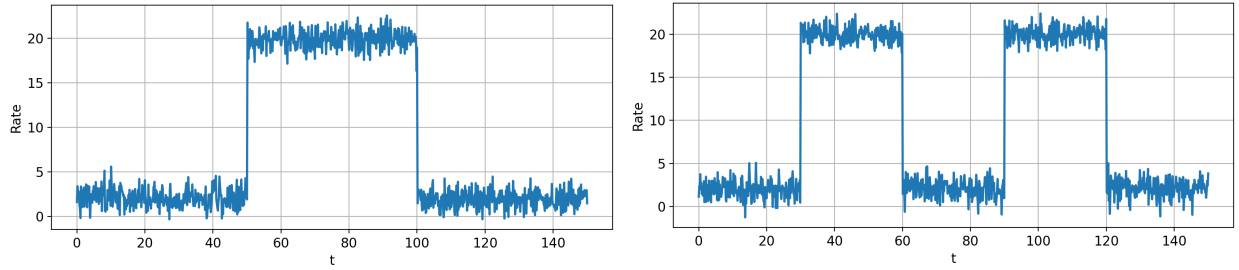


Figure 2 Left: single congestion shock. Right: two congestion shocks. Predicted arrival rate $\hat{\lambda}(t | 0) = \lambda(t) + \epsilon_{0,t}$, where $\epsilon_{0,t} \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 1$.

Figure 3 compares the performance of the benchmark and look-ahead policies by plotting the trade-off between total patient waiting time and the proportion of rejected transfer requests. Across both panels, corresponding to review intervals $\delta = 1$ and $\delta = 5$ respectively, the look-ahead policy uniformly dominates the benchmark. Specifically, for any given level of waiting time, the look-ahead policy achieves a lower rejection rate, and conversely, for any given rejection rate, it yields shorter waiting times. These results highlight the operational value of incorporating predictive demand information into dynamic admission decisions, particularly in environments subject to transient congestion shocks.

5.1.2. Two congestion shocks We next consider two congestion shocks, with $\lambda(t) = 20$ for $t \in (30, 60] \cup (90, 120]$, $\lambda(t) = 2$ otherwise on $[0, T]$. The right panel of Figure 2 illustrates a realization of the predicted arrival rate $\hat{\lambda}(t | 0)$.

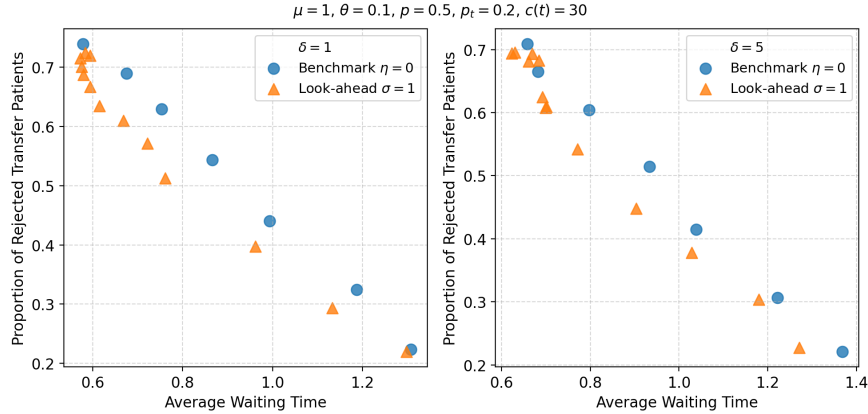


Figure 3 Single congestion shock: comparison between look-ahead v.s. benchmark.

Figure 4 compares the performance of the benchmark and look-ahead policies by plotting the trade-off between total patient waiting time and the fraction of rejected transfer requests. For both review intervals $\delta = 1$ and $\delta = 5$, the look-ahead policy again consistently dominates the benchmark. This demonstrates that the benefits of predictive admission control persist even in more complex environments with multiple congestion surges and intervening recovery periods.

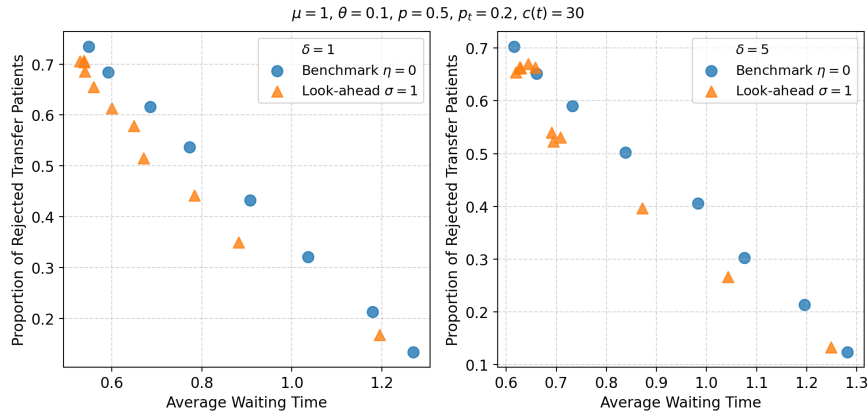


Figure 4 Two congestion shocks: comparison between look-ahead v.s. benchmark.

5.2. Prediction Errors

In Section 5.1, we evaluated the policy under a baseline forecast-error model with unbiased, independently distributed noise in the predicted arrival rates. We now examine robustness to a broader range of forecast misspecification.

Rather than deriving forecast errors from a particular prediction algorithm, we specify the error process $e_\lambda(s | t) = \hat{\lambda}(s | t) - \lambda(s)$ directly. This allows us to vary distinct dimensions of forecast quality one at a time, including noise magnitude, temporal persistence, horizon dependence, systematic bias, and timing error, and isolate their effects on admission-control performance. In this way, we can assess the operational consequences of different forms of forecast error without confounding them with the choice of a particular prediction method. Accordingly, the experiments in

this section should be interpreted as controlled forecast-error stress tests rather than as an evaluation of any specific forecasting algorithm. Section 6 complements this analysis with an end-to-end implementation in which forecasts are generated by models trained and evaluated on hospital data.

Recall that for each review epoch t , $s_k = t + k\Delta t$, $k = 0, 1, \dots$, is the future forecast grid. We specify the forecast-error process $e_\lambda(s_k | t)$ directly.

Scenario I: *Independent forecast errors (baseline).*

$e_\lambda(s_k | t) \sim \mathcal{N}(0, \sigma^2)$. We consider $\sigma \in \{0, 1, 2, 4\}$ to assess sensitivity to the noise level.

Scenario II: *Correlated forecast errors.*

$e_\lambda(s_k | t) = \rho e_\lambda(s_{k-1} | t) + u_{t,k}$, $u_{t,k} \sim \mathcal{N}(0, \sigma_u^2)$, where $\rho \in \{0.7, 0.99\}$ and $\sigma_u^2 = (1 - \rho^2)\sigma^2$.

Scenario III: *Horizon-dependent forecast uncertainty.*

$e_\lambda(s_k | t) \sim \mathcal{N}(0, \sigma^2(1 + \beta k))$. We consider $\beta \in \{0.02, 0.2\}$.

Scenario IV: *Biased forecasts.*

$e_\lambda(s_k | t) = \alpha \lambda(s_k) + \epsilon_{t,k}$, $\epsilon_{t,k} \sim \mathcal{N}(0, \sigma^2)$, i.e., $\hat{\lambda}(s_k | t) = (1 + \alpha)\lambda(s_k) + \epsilon_{t,k}$. We consider $\alpha \in \{-0.6, -0.4, -0.2, 0.2, 0.4, 0.6\}$. Positive values of α correspond to persistent overestimation of demand, while negative values represent underestimation.

Scenario V: *Shock timing misprediction.*

$e_\lambda(s_k | t) = \lambda(s_k - \xi) - \lambda(s_k) + \epsilon_{t,k}$, $\epsilon_{t,k} \sim \mathcal{N}(0, \sigma^2)$, i.e., $\hat{\lambda}(s_k | t) = \lambda(s_k - \xi) + \epsilon_{t,k}$. We consider $\xi \in \{5, 10\}$.

For Scenarios II–V, we consider $\sigma \in \{1, 2\}$. For ease of exposition, the main paper reports results for the two-surge scenario with $\sigma = 2$ and $\delta = 1$. Results for additional cases, including the single-surge scenario and alternative σ and δ , are qualitatively similar and reported in Appendix D.

Across the baseline specification (Scenario I) and Scenarios II–IV with positive α , the performance of the look-ahead policy remains qualitatively stable. As shown in Figure 5, we observe little systematic deterioration in the trade-off between patient waiting and transfer rejection under unbiased noise, persistent (correlated) forecast errors, horizon-dependent uncertainty, or systematic overprediction of demand. These results suggest that the policy is relatively insensitive to forecast errors that perturb the predicted rate trajectory without substantially obscuring the magnitude or timing of impending congestion.

In contrast, performance deteriorates under systematic underprediction of demand (Scenario IV with negative α) and errors in surge timing (Scenario V), as illustrated in Figure 6. Underprediction causes the policy to underestimate the severity of future congestion, while delayed surge predictions prevent it from responding sufficiently early. These results highlight an important asymmetry in the operational consequences of forecast errors: errors that obscure the magnitude or timing of impending congestion are substantially more consequential than moderate noise that preserves these features of the future demand trajectory.

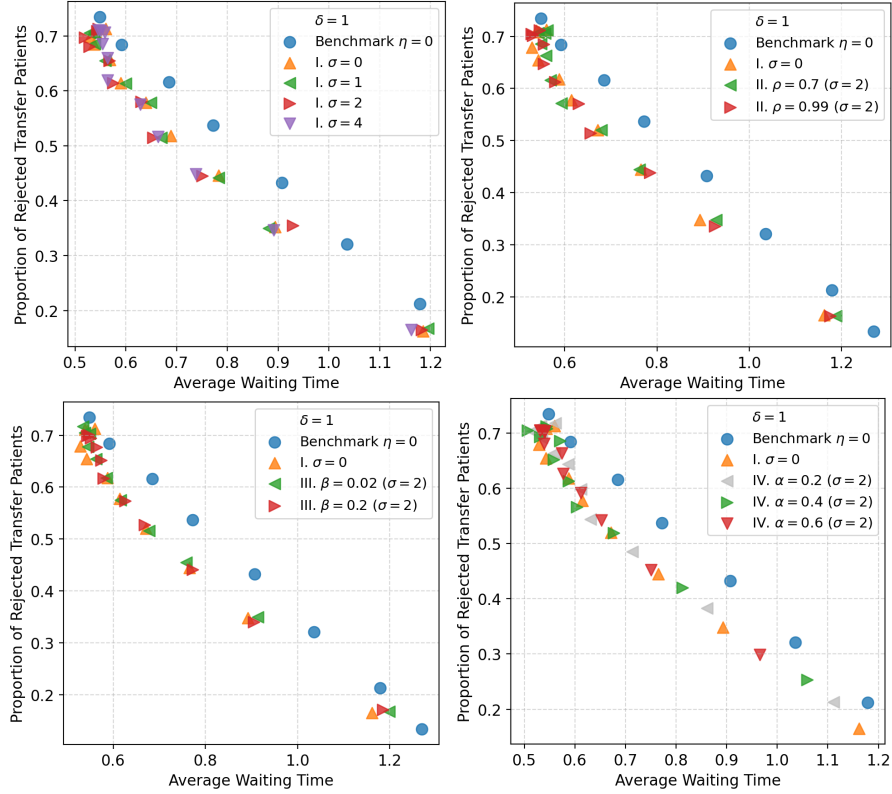


Figure 5 Similar performance with prediction errors in scenarios I, II, III, IV with positive α

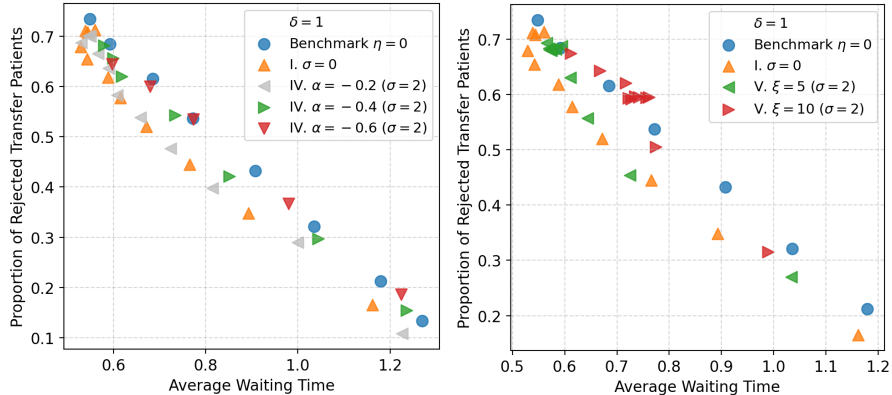


Figure 6 Degradation in performance with prediction errors in scenarios IV (with negative α) and V

6. Case Study: Implementation with Real Hospital Data

In this section, we evaluate the proposed look-ahead admission control policy using operational data from a large academic medical center. The purpose of this case study is to assess whether the policy can be implemented with realistically available information, including data-driven forecasts of demand and boarding delays, and whether it yields measurable improvements relative to the hospital's current transfer-admission practice when calibrated to empirical operating conditions.

A central practical consideration is that transfer patients differ systematically from non-transfer patients in their likelihood of requiring inpatient admission. To capture this empirically important

heterogeneity while preserving the policy logic developed earlier, Section 6.1 generalizes the base model to allow type-dependent admission probabilities. This extension modifies only the fluid dynamics used to compute the hitting time; the admission decision remains a threshold rule in the predicted hitting time.

Building on this implementation model, Section 6.2 describes the data sources, forecasting models for arrivals and processing rates, and parameter calibration. Section 6.3 compares the performance of the look-ahead policy to the hospital's current policy using simulated counterfactuals over the study period. Together, the results illustrate how predictive information can be operationalized through a periodic-review rule by quantifying the improvements in the congestion-access trade-off.

6.1. Operational Model with Heterogeneous Admission Outcomes

To support implementation, we generalize the base model by allowing the probability of inpatient admission to depend on patient type. Specifically, patients are classified into external patients (type 1) and transfer patients (type 2). Let p_1 and p_2 denote the probabilities that type-1 and type-2 patients, respectively, are discharged directly after completing ED service. This distinction matters because transfer patients are typically more likely to require inpatient admission.

The stochastic primitives are as in Section 2; in particular, arrivals, service times, abandonment behavior, and boarding dynamics are unchanged. We additionally assume that transfer patients are prioritized for ED service whenever capacity is available, which reflects common operational practice when transfer patients arrive through coordinated pathways.

For implementation, we work with the corresponding fluid approximation. Let $x_i(t)$, $z_i(t)$, and $q_i(t)$ denote the fluid analogues of the number of type- i patients in system, in service, and waiting in queue, respectively, where $q_i(t) = x_i(t) - z_i(t)$, for $i \in \{1, 2\}$. Let $b(t)$ denote the boarding fluid. The fluid dynamics are

$$\begin{aligned} \dot{x}_1(t) &= \lambda_1(t) - \mu z_1(t) - \theta q_1(t), \quad \dot{x}_2(t) = g(t)\lambda_2(t) - \mu z_2(t) - \theta q_2(t), \\ \dot{b}(t) &= (1 - p_1)\mu z_1(t) + (1 - p_2)\mu z_2(t) - \nu(t)b(t), \end{aligned} \quad (11)$$

where $g(t) \in [0, 1]$ denotes the (fluid) transfer acceptance control. Service capacity is allocated according to transfer-priority:

$$z_2(t) = \min\{c(t) - b(t), x_2(t)\}, \quad z_1(t) = \min\{c(t) - b(t) - z_2(t), x_1(t)\}. \quad (12)$$

We retain the hitting-time structure of the policy in Section 3 as a heuristic for this heterogeneous model. Let $H_t(x_1, x_2, b)$ denote the time until both fluid queues clear, starting from (x_1, x_2, b) at time t under full acceptance $g(s) \equiv 1$ for $s \geq t$. We implement a periodic-review policy: at review time $k\delta$, we compute $\hat{H}_{k\delta}$ using the current state $(X_1(k\delta), X_2(k\delta), B(k\delta))$ and the forecasted rates, and hold the admission decision fixed throughout $[k\delta, (k+1)\delta)$. Specifically, transfer requests are blocked during this interval if

$$\hat{H}_{k\delta}(X_1(k\delta), X_2(k\delta), B(k\delta)) - \frac{\delta}{2} > -\frac{1}{\theta} \log\left(1 - \frac{l\theta}{h}\right), \quad (13)$$

and accepted otherwise. The shift term $\delta/2$ is a simple correction for within-window state evolution under periodic review.

6.2. Data, Prediction, and Model Calibration

We conduct the case study using detailed ED patient flow data, including some inter-facility transfer requests, from April 2022 to November 2024 at a large quaternary-care academic ED in the northeastern United States. The data contain time-stamped records of ED arrival, physician first contact, inpatient bed request, and ED departure, which allow us to reconstruct time-varying arrival patterns, service dynamics, and boarding delays. To support forecasting and calibration, we additionally collect and construct time-varying measures of inpatient occupancy (including non-ED admissions), extract the hospital’s recorded transfer-admission status, and merge local weather variables (temperature, precipitation, snowfall, and wind speed). Additional details on data processing are provided in Appendix C.1.

The resulting dataset contains 311,739 ED encounters (301,586 external arrivals, 6,826 completed ED-to-ED transfers, and 3,327 rejected ED-to-ED transfers), of which 21.04% involve a request for inpatient admission. We define boarding time as the duration between inpatient bed request and ED departure. On average, the ED experiences 13.32 arrivals per hour. Boarding times have a mean of 17.88 hours, and their empirical distribution (Figure 7) exhibits substantial dispersion and a pronounced right tail.

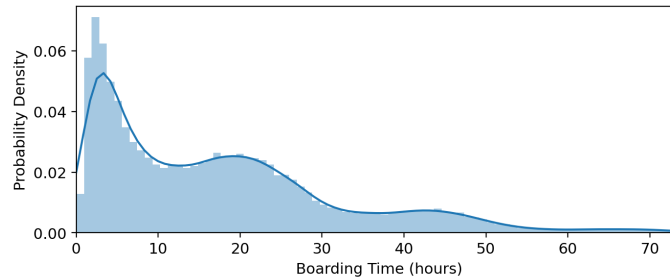


Figure 7 Histogram of boarding time in hours

We first instantiate the forecasting framework of Section 4.2 using statistical and machine-learning models trained on historical hospital data.

Arrival-rate prediction. We forecast arrivals at a two-hour resolution using information available at the beginning of each interval (see Appendix C.3 for details of arrival patterns). The predictors include: (i) calendar variables (month, day-of-week, time-of-day, holiday indicators); (ii) lagged arrival counts over the preceding one to seven intervals; and (iii) weather variables (temperature, precipitation, snowfall, wind, and indicators for extreme heat). For multi-step prediction, we use realized weather as a proxy for short-term forecasts available at the review epoch. To avoid look-ahead bias, we split the data chronologically into equally sized training and test samples and fit

both linear regression and boosted tree models (estimated with XGBoost). Table 1 reports their predictive accuracy; both models achieve similar test-set RMSE (5.69 for linear regression and 5.66 for XGBoost), suggesting similar out-of-sample prediction accuracy.

Dataset	Linear Regression	XGBoost
Train	5.62	4.29
Test	5.69	5.66

Table 1 RMSE values for arrival count prediction models.

Processing-rate estimation and prediction. We next estimate and predict the processing rate governing transfers from the ED to inpatient units. We assume boarding times are exponential and the processing rate $\nu(t)$ is piecewise constant over prespecified 8-hour intervals, partitioning each 24-hour day into three 8-hour periods. For a given interval $[t_k, t_{k+1})$, we include all patients who spend any time boarding during that interval. Each patient contributes exposure from the later of their boarding start time and t_k until either (i) the patient completes boarding and leaves the ED for an inpatient unit within the interval, or (ii) the interval ends, in which case the observation is right-censored at t_{k+1} . Thus, patients who complete boarding contribute both a transfer-completion event and their time at risk, whereas patients who remain boarding at the end of the interval contribute only their accumulated time at risk. Under the exponential assumption, the likelihood therefore depends on the number of completed transfers and the total boarding exposure accumulated during the interval. Maximizing this likelihood yields the closed-form estimator $\nu_k = (\text{\#completions})/(\text{total exposure})$. We then set $\nu(t) = \nu_k$ for $t \in [t_k, t_{k+1})$ (see Appendix C.2 for details). The average estimated processing rate is 0.06.

We then use the interval-specific estimates ν_k as the response variable for forecasting future inpatient processing rates. Predictors include temporal indicators, lagged estimate of the processing rate, and weather variables. As with the arrival rate forecast, we split the data chronologically into equally sized training and test periods to preserve the temporal ordering of the observations, and fit both linear regression and XGBoost models on the training sample. Table 2 reports resulting predictive accuracy. The two models achieve similar out-of-sample performance, with test-set RMSEs of approximately 0.02.

Dataset	Linear Regression	XGBoost
Train	0.02	0.01
Test	0.02	0.02

Table 2 RMSE values for processing rate prediction models.

Model calibration and simulation. We calibrate a multi-server queueing simulator to reproduce key features of observed ED operations over the study period. Arrivals follow a time-varying Poisson process with piecewise-constant rates over two-hour intervals, estimated directly from the data. The observed fraction of arrivals classified as transfer patients is $\gamma = 0.033$. However, discussions with our clinical partners suggest that transfer requests may be substantially underrecorded in the

administrative data (e.g., declined transfers not systematically entered into the transfer data logs). Therefore, we treat $\gamma = 0.033$ as a conservative lower-bound scenario and additionally consider larger values of γ to assess settings with greater transfer exposure consistent with rates observed elsewhere (Moore et al. 2023). Patient patience times are assumed to be exponentially distributed, with abandonment rate estimated by maximum likelihood as $\theta = 0.0347$. Service times are also assumed exponential, with an estimated mean of 8.1620 hours, corresponding to a service rate $\mu = 1/8.1620$. We set the number of servers to $c(t) = 153$, equal to the median number of patients simultaneously in service during the study period. The probability of direct discharge after ED service is estimated separately for external and transfer patients, yielding $p_1 = 0.7928$ and $p_2 = 0.5467$, respectively. Patients who are not discharged directly enter the boarding stage, with transfer to inpatient units governed by the estimated time-varying processing rate $\nu(t)$. To improve numerical stability while preserving the empirical tail behavior, we truncate service and boarding times at the 99.5th percentiles of their observed distributions (84 and 72 hours, respectively).

In implementing the look-ahead policy, we compute the hitting-time functional using recursively generated forecasts of the future arrival and inpatient processing rates. In particular, at each review epoch t , the trained models produce forecast trajectories $\hat{\lambda}(s | t)$ and $\hat{\nu}(s | t)$, $s \geq t$, using only information available up to time t . Because both forecasting models include lagged response variables, multi-step forecasts are generated recursively: predicted values from earlier forecast steps are used as lagged inputs when constructing predictions for subsequent periods. The resulting forecast trajectories are then used to evaluate the hitting-time functional. For the two arrival streams, we set $\hat{\lambda}_1(s | t) = (1 - \gamma)\hat{\lambda}(s | t)$, $\hat{\lambda}_2(s | t) = \gamma\hat{\lambda}(s | t)$. We vary the cost ratio l/h over its feasible range $0 < l/h < 1/\theta$ and consider review intervals δ ranging from one hour to twelve hours. The parameter grid considered is

$$\frac{l}{h} \in \{0.5, 1, 4, 8, 12, 16, 20, 24, 26\}, \delta \in \{1, 2, 4, 6, 8, 12\} \text{ hours}, \gamma \in \{0.0331, 0.1, 0.2\}. \quad (14)$$

For each configuration, we report the fraction of rejected transfer requests, average waiting time, and average boarding census. To mitigate transient effects, we compute performance after excluding the first and last months of each simulated period.

6.3. Performance Evaluation

The hospital's current policy blocks all transfer requests whenever the boarding census exceeds a fixed threshold $\bar{b}$. During the study period, $\bar{b}$ is approximately 45, and the blocking indicator is active 81.04% of the time. To benchmark this policy, we simulate the system under thresholds $\bar{b} \in \{41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 65\}$. When $\bar{b} = 45$ with $\delta = 1$, the simulated fractions of time that transfers are blocked are 82.33%, 82.27%, and 77.86% for $\gamma = 3.31\%$, 10%, and 20%, respectively, close to the observed blocking frequency of 81.04%. Note that these time-based blocking

measures differ from the fraction of transfer requests rejected reported below because the transfer arrival rate varies over time.

We evaluate the look-ahead policy (13) over the parameter grid in (14). Under the hospital's boarding-threshold policy, increasing $\bar{b}$ reduces transfer rejection at the expense of longer patient waiting times. Increasing the cost ratio l/h under the look-ahead policy produces a similar trade-off. This congestion–access trade-off is less pronounced when $\gamma = 3.31\%$, because transfers account for a small share of arrivals and rejecting them has limited impact on aggregate congestion. Detailed results for each policy, including boarding census and sensitivity to the review interval, are reported in Appendix C.4.

Figure 8 compares the congestion–access trade-offs achieved by the two policies, using a one-hour review interval for the look-ahead policy. For both $\gamma = 0.10$ and $\gamma = 0.20$, the look-ahead policy achieves a more favorable congestion–access trade-off, yielding lower transfer rejection rates for comparable or shorter patient waiting times. For example, when $\gamma = 0.20$ and the average waiting time is capped at 2 hours, the look-ahead policy (with $l/h = 12$) reduces the rejection fraction from 0.53 (under the policy $\bar{b} = 57$) to 0.48, corresponding to a 9% reduction in declines (equivalently, 11% increase in acceptances). Conversely, when the rejection fraction is constrained to be no more than 0.50, the look-ahead policy reduces average waiting time from 2.11 hours under the hospital policy ($\bar{b} = 59$) to 1.89 hours, a reduction of approximately 10%. These comparisons demonstrate that incorporating forecasts of future operating conditions into a simple periodic-review rule can meaningfully improve the congestion–access trade-off relative to current practice.

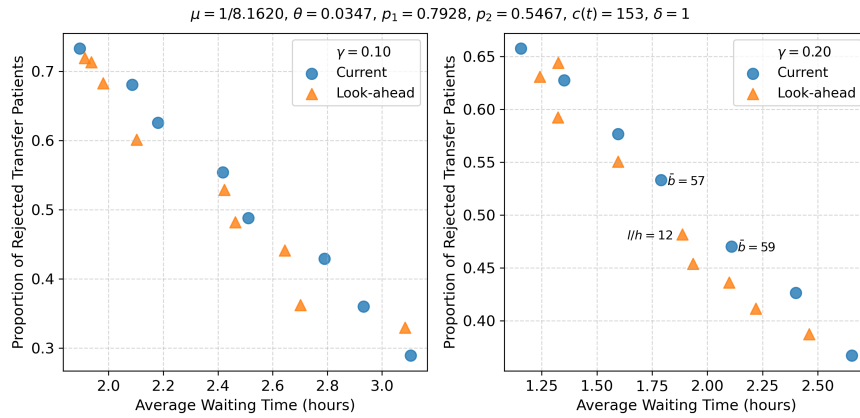


Figure 8 Comparison between Look-ahead Policy versus Current Policy

Additional forecasting specification. Appendix C.5 examines an expanded forecasting specification that adds ED census to the arrival-rate model and boarding census and service-level inpatient occupancy to the processing-rate model. In this exercise, we use the realized future values of these additional predictors over the look-ahead horizon. Because these values are unavailable at the review epoch, an operational implementation would require additional models to forecast them.

This exercise therefore serves as a sensitivity analysis with additional information, rather than an implementable forecasting specification. Predictive accuracy and policy performance are similar to those under the main specification, which uses only information available at each review epoch.

7. Conclusion

We develop a proactive admission control framework for emergency care settings that integrates predictive information on patient arrivals and inpatient admission delays. As crowding increasingly threatens care quality and contributes to staff burnout, our approach provides a practical, data-driven mechanism for improving patient flow and system performance.

To address the operational complexity of admission decisions under transient congestion, we adopt a fluid approximation and characterize the optimal fluid control. Guided by this analysis, we propose a look-ahead admission policy that is intuitive, easy to implement, and asymptotically optimal as demand and capacity scale.

Extensive synthetic experiments and a data-calibrated case study based on real hospital data demonstrate the effectiveness of the proposed approach. Across a wide range of operating conditions, the look-ahead policy consistently outperforms non-predictive benchmarks by achieving better trade-offs between patient waiting times and transfer rejections, enabling hospitals to improve patient access (i.e., decrease transfer rejections) while maintaining waiting times.

The magnitude of these benefits depends on both transfer-request volume and congestion. The policy is particularly relevant to tertiary and referral hospitals that receive substantial transfer demand and experience severe ED crowding. When transfer requests account for a small share of demand, the potential reduction in aggregate ED congestion is more limited, although managing these requests may still be useful during periods of acute crowding.

Several directions merit further investigation. First, the framework could be extended to jointly manage inbound and outbound transfers. Although outbound ED-to-ED transfers are rare at our study hospital, which offers a broad range of clinical services, they may provide an additional means of managing congestion at other hospitals. Direct ED-to-inpatient transfers offer another option: patients who have completed their ED evaluation and require admission may be transferred directly to an available inpatient bed at another hospital. When treated as exogenous, these departures can be incorporated into the boarding processing rate $\nu(t)$. Explicitly controlling them would require accounting for bed availability at receiving hospitals, transfer costs, and clinical eligibility. A network extension could examine how hospitals coordinate these decisions to improve access and reduce congestion across the system.

Second, a prospective pilot would help assess the operational value of the proposed policy. Our approach is designed to facilitate implementation: it combines forecasts of arrival and inpatient processing rates with an interpretable decision rule applied at regular review intervals. Its cost

ratio provides a transparent way to adjust the balance between congestion and transfer access. A good evaluation would help identify appropriate review intervals and procedures for updating forecasts as operating conditions change.

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Online Appendix to Proactive Transfer Admission Control for Emergency Departments

Appendix A: Proof of the Fluid Optimal Control (Theorems 1–2)

In this section, we prove the optimal fluid admission control policy via Pontryagin's Minimum Principle. We first state a version of Pontryagin's Minimum Principle adapted to (5). We then verify the (sufficient) conditions of Pontryagin's Minimum Principle.

Fix any admissible $g(\cdot)$. It is without loss of optimality to enforce work conservation, which implies the service rate $z(t)$ must satisfy: $z(t) = \min\{x(t), c(t) - b(t)\}$. Equivalently, this can be formulated as treating $z(t)$ as an auxiliary variable subject to the constraints: $z(t) \leq x(t)$, $z(t) \leq c(t) - b(t)$, and $z(t) = c(t) - b(t)$ if $x(t) > c(t) - b(t)$. For the purpose of deriving the optimal control, we define the **Hamiltonian** $\mathcal{H}$ representing the system dynamics and costs, and the **Augmented Lagrangian** $\mathcal{L}$ which incorporates the boundary constraints via multipliers.

Define the Hamiltonian $\mathcal{H} : [0, \tau] \times \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}$ as: $\mathcal{H}(t, x, b, g, p_x, p_b) := p_x \left[\lambda_1(t) + g\lambda_2(t) - \mu z - \theta(x - z) \right] + p_b \left[(1 - p)\mu z - \nu(t)b \right] + h(x - z) + l\lambda_2(t)(1 - g)$. To handle the nonsmooth dynamics $z = \min\{x, c - b\}$, we introduce the Lagrangian multipliers $\eta_1(t), \eta_2(t) \geq 0$ corresponding to the capacity and queue constraints, respectively. The Augmented Lagrangian is: $\mathcal{L}(t, x, b, g, z, p, \eta) := \mathcal{H} - \eta_1(c(t) - b - z) - \eta_2(x - z)$.

The following sufficiency theorem is adapted from standard results on optimal control with state constraints (e.g., (Seierstad and Sydsæter 1987, Hartl et al. 1995)).

THEOREM 3. *Consider the optimal control problem (5). Let (x^*, b^*, g^*, z^*) be a feasible trajectory. Suppose there exist absolutely continuous adjoint functions $p_x(\cdot), p_b(\cdot)$ and piecewise continuous multipliers $\eta_1(\cdot), \eta_2(\cdot) \geq 0$ defined on $[0, \tau]$ such that for almost all t :*

1. **Convexity:** *The map $(x, b, g) \mapsto \mathcal{H}(t, x, b, g, p_x, p_b)$ is convex on $\mathbb{R}^2 \times [0, 1]$.*
2. **Adjoint Equations:** *The time derivatives of the adjoint variables satisfy: $-\dot{p}_x(t) = \frac{\partial \mathcal{L}}{\partial x}$, $-\dot{p}_b(t) = \frac{\partial \mathcal{L}}{\partial b}$, evaluated at the optimal solution.*
3. **Stationarity & Complementarity:** *The auxiliary variable z^* maximizes the Lagrangian, and the multipliers satisfy complementary slackness: $\frac{\partial \mathcal{L}}{\partial z} = 0, \eta_1(c - b^* - z^*) = 0, \eta_2(x^* - z^*) = 0$.*
4. **Minimization Condition:** $g^*(t) \in \arg \min_{0 \leq g \leq 1} \mathcal{H}(t, x^*(t), b^*(t), g, p_x(t), p_b(t))$.
5. **Transversality:** $p_x(\tau) = 0, p_b(\tau) = 0$.

Then $(x^(\cdot), b^*(\cdot), g^*(\cdot))$ is an optimal solution.*

REMARK 1. We do not use the global augmented Lagrangian in our proofs. The reason is that, once the trajectory is decomposed into congested and uncongested phases, the active relation for $z(t)$ is known in each phase (namely, $z(t) = c(t) - b(t)$ when $q(t) > 0$ and $z(t) = x(t)$ when $q(t) = 0$), so the dynamics become smooth after substitution. Therefore, the optimality conditions can be verified using standard phasewise Hamiltonians, with the phases patched together through continuity of the state and costate at the switching times.

A.1. Proof of Theorem 1

We prove the theorem by a verification argument based on Pontryagin's minimum principle on the congested interval.

Under Assumption 1, the fluid queue remains positive before the clearing time and empties at τ ; in particular, $q(t) > 0$ for $t \in [0, \tau)$ and $q(\tau) = 0$. Hence, on $[0, \tau)$, $z(t) = c(t) - b(t)$. Define the Hamiltonian (with $z = c - b$ already substituted) by $\mathcal{H}_c(t, x, b, g, p_x, p_b) := p_x [\lambda_1(t) + g\lambda_2(t) - \mu(c(t) - b) - \theta(x + b - c(t))] + p_b [(1 - p)\mu(c(t) - b) - \nu(t)b] + h(x + b - c(t)) + l\lambda_2(t)(1 - g)$. The adjoint equations on $[0, \tau)$ are $-\dot{p}_x(t) = \frac{\partial \mathcal{H}_c}{\partial x} = -\theta p_x(t) + h$, $-\dot{p}_b(t) = \frac{\partial \mathcal{H}_c}{\partial b} = (\mu - \theta)p_x(t) - ((1 - p)\mu + \nu(t))p_b(t) + h$. Using the transversality conditions $p_x(\tau) = 0, p_b(\tau) = 0$ (since the terminal cost is zero at the clearing time), we have

$$p_x(t) = \frac{h}{\theta} (1 - e^{-\theta(\tau-t)}), \quad t \in [0, \tau]. \quad (15)$$

(When $\theta = 0$, the same calculation yields $p_x(t) = h(\tau - t)$.)

Next, since $\frac{\partial \mathcal{H}_c}{\partial g}(t) = \lambda_2(t)(p_x(t) - l)$, pointwise minimization of the Hamiltonian yields $g^*(t) \in \arg \min_{0 \leq g \leq 1} \mathcal{H}_c(t, x, b, g, p_x, p_b) = 1$ when $p_x(t) < l$, $g^*(t) = 0$ when $p_x(t) > l$, and $g^*(t) \in [0, 1]$ when $p_x(t) = l$. Therefore, any optimal control must satisfy the threshold rule

$$g^*(t) = \mathbf{1}\{p_x(t) \leq l\} \quad \text{for a.e. } t \in [0, \tau], \quad (16)$$

where the value on the measure-zero set $\{t : p_x(t) = l\}$ can be chosen arbitrarily.

Recall that $H_t(x, b)$ is defined as the time-to-clear under the *all-accept* policy $g \equiv 1$, starting from state (x, b) at time t . We now show that along the optimal trajectory, the sign of $p_x(t) - l$ is exactly the sign of $\frac{h}{\theta}(1 - e^{-\theta H_t(x(t), b(t))}) - l$. Define $K := \inf \{u \geq 0 : \frac{h}{\theta}(1 - e^{-\theta u}) \geq l\}$. Equivalently, if $l < h/\theta$, then $K = -\frac{1}{\theta} \log(1 - \frac{l\theta}{h})$, while if $l \geq h/\theta$, then $K = +\infty$ and the policy is trivially $g^* \equiv 1$ (because $p_x(t) \leq h/\theta \leq l$ for all t by (15)). Henceforth, assume $l < h/\theta$, so that $K < \infty$. By (15), $p_x(t)$ is strictly decreasing in t and crosses the level l exactly once at time $\zeta := \tau - K$. Consequently, $g^*(t) = 0$ for $t < \zeta$ and $g^*(t) = 1$ for $t > \zeta$ a.e., with an arbitrary choice at $t = \zeta$.

Phase A: $t \in [\zeta, \tau]$. The optimal policy accepts all transfers from time t onward, i.e., $g^*(s) = 1$ for all $s \in [t, \tau]$. Therefore, the actual trajectory on $[t, \tau]$ coincides with the trajectory used in the definition of the hitting time $H_t(x(t), b(t))$. Since the queue clears at time τ , we obtain $H_t(x(t), b(t)) = \tau - t$, $t \in [\zeta, \tau]$. Together with (15), this gives $p_x(t) = \frac{h}{\theta}(1 - e^{-\theta H_t(x(t), b(t))})$, $t \in [\zeta, \tau]$. Hence on Phase A, $p_x(t) \leq l \iff \frac{h}{\theta}(1 - e^{-\theta H_t(x(t), b(t))}) \leq l$, and thus (7) agrees with (16).

Phase B: $t \in [0, \zeta)$. At such a time, the optimal policy rejects transfers until ζ , whereas the hitting-time function $H_t(x(t), b(t))$ is computed under the all-accept policy $g \equiv 1$. Rejecting transfers cannot delay clearing relative to accepting all transfers; therefore the all-accept clearing time from $(x(t), b(t))$ is no smaller than the actual remaining clearing time under g^* : $H_t(x(t), b(t)) \geq \tau - t > \tau - \zeta = K$, $t \in [0, \zeta)$. (The strict inequality $\tau - t > K$ follows from $t < \zeta = \tau - K$.) This further implies $\frac{h}{\theta}(1 - e^{-\theta H_t(x(t), b(t))}) > \frac{h}{\theta}(1 - e^{-\theta K}) = l$. Thus on Phase B, $\frac{h}{\theta}(1 - e^{-\theta H_t(x(t), b(t))}) > l$, which is equivalent to $g^*(t) = 0$ in (7), again agreeing with (16).

Combining Phases A and B, we conclude that the pointwise Hamiltonian minimizer (16) can be written exactly as $g^*(t) = \mathbf{1}\left\{\frac{h}{\theta}(1 - \exp(-\theta H_t(x(t), b(t)))) \leq l\right\}$, which is the claimed policy.

Overall, by construction, the state trajectory under g^* is feasible for (5). The adjoint pair (p_x, p_b) satisfies the adjoint equations with transversality, and g^* minimizes the Hamiltonian pointwise on $[0, \tau]$ by (16). Therefore, by the verification theorem stated in Theorem 3 (applied on the congested interval $[0, \tau]$, where $z = c - b$ and the dynamics are smooth), the control g^* is optimal for (5). $\square$

A.2. Proof of Theorem 2

Recall that $I_1 = [0, \tau_1]$ and $I_2 = [\sigma_2, \tau_2]$ are the two congestion periods, separated by the uncongested gap $G_1 = [\tau_1, \sigma_2]$. Note that if G_1 is empty, the system effectively behaves as a single congested interval, and the proof of Theorem 1 applies directly. we therefore focus on the case where G_1 is non-empty.

The proof constructs adjoint variables via backward integration from τ_2 (transversality: $p_x(\tau_2) = p_b(\tau_2) = 0$) and verifies Theorem 3 in three phases: Phase 1, first congested period, $t \in [0, \tau_1]$; Phase 2, uncongested period, $t \in [\tau_1, \sigma_1]$; Phase 3, second congested period, $t \in [\sigma_1, \tau_2]$.

Costate continuity at regime-switching times. In state-constrained optimal control, the costate may in general admit jumps at times when the trajectory enters or leaves the boundary of the state constraint. In our problem, the relevant switching surface is $q(t) = x(t) + b(t) - c(t) = 0$. For the proof below, we use a phasewise Hamiltonian formulation and select the costate to be continuous at the switching times τ_1 and σ_1 . This is consistent with the standard jump conditions for first-order state constraints (see, e.g., Hartl et al. 1995): the costate jump at a switching time t_s has the form $p(t_s^-) - p(t_s^+) = \pi_s \nabla_x q(t_s)$, for a scalar junction multiplier π_s . Under our assumptions, the trajectory crosses the boundary $q = 0$ transversally (i.e., with nonzero crossing speed $\dot{q}(t_s) \neq 0$), and there is no boundary/impulse cost at the switching time. In this regular crossing case, the junction multiplier vanishes, $\pi_s = 0$, so the costate is continuous: $p(t_s^-) = p(t_s^+)$, $t_s \in \{\tau_1, \sigma_1\}$ (Equivalently, one may view the phasewise Hamiltonian formulation as a specialization of the augmented-Lagrangian formulation with a multiplier choice for which the costate is continuous across regime switches.)

This continuity allows us to propagate, backward in time, the ending point of one phase as the starting point of the preceding phase.

Phase 3 ($t \in [\sigma_1, \tau_2]$): Second Congested Period. Since $q^*(t) = x^*(t) + b^*(t) - c(t) > 0$, the service rate is $z^*(t) = c(t) - b^*(t)$. The congested-phase Hamiltonian is $\mathcal{H}_c(t, x, b, g, p_x, p_b) := p_x [\lambda_1(t) + g\lambda_2(t) - \mu(c(t) - b) - \theta(x + b - c(t))] + p_b [(1-p)\mu(c(t) - b) - \nu(t)b] + h(x + b - c(t)) + l\lambda_2(t)(1-g)$. The adjoint equations are $-\dot{p}_x(t) = \frac{\partial \mathcal{H}_c}{\partial x} = -\theta p_x(t) + h$, $-\dot{p}_b(t) = \frac{\partial \mathcal{H}_c}{\partial b} = (\mu - \theta)p_x(t) - ((1-p)\mu + \nu(t))p_b(t) + h$. With $p_x(\tau_2) = 0$, we obtain $p_x(t) = \frac{h}{\theta} (1 - e^{-\theta(\tau-t)})$. For p_b , let $\beta(t) = (1-p)\mu + \nu(t)$. With $p_x(\tau_2) = p_b(\tau_2) = 0$, we have

$$p_b(t) = \int_t^{\tau_2} [h + (\mu - \theta)p_x(s)] \exp\left(-\int_t^s \beta(u)du\right) ds. \quad (17)$$

Phase 2 ($t \in [\tau_1, \sigma_1]$): Uncongested Period. In this phase, $q^*(t) = 0$ and $z^*(t) = x^*(t)$. The uncongested-phase Hamiltonian is therefore $\mathcal{H}_u(t, x, b, g, p_x, p_b) := p_x [\lambda_1(t) + g\lambda_2(t) - \mu x] + p_b [(1-p)\mu x - \nu(t)b] + l\lambda_2(t)(1-g)$. The adjoint equations are

$$-\dot{p}_x(t) = \frac{\partial \mathcal{H}_u}{\partial x} = -\mu p_x(t) + (1-p)\mu p_b(t), \quad (18)$$

$$-\dot{p}_b(t) = \frac{\partial \mathcal{H}_u}{\partial b} = -\nu(t)p_b(t). \quad (19)$$

The boundary values at $t = \sigma_1$ are inherited continuously from Phase 3. In particular, integrating (19) backward from σ_1 , we have $p_b(t) = p_b(\sigma_1) \exp\left(-\int_t^{\sigma_1} \nu(u) du\right)$. Integrating (18) backward from σ_1 , we have $p_x(t) = e^{-\mu(\sigma_2-t)} p_x(\sigma_2) + (1-p)\mu \int_t^{\sigma_2} e^{-\mu(u-t)} \exp\left(-\int_u^{\sigma_2} \nu(r) dr\right) du p_b(\sigma_2)$.

Phase 1 ($t \in [0, \tau_1]$): First Congested Period. As in Phase 3, the system is congested and the Hamiltonian is $\mathcal{H}_c$. Thus the adjoint equations again satisfy $-\dot{p}_x(t) = -\theta p_x(t) + h$, $-\dot{p}_b(t) = (\mu - \theta)p_x(t) - ((1-p)\mu + \nu(t))p_b(t) + h$. The values at $t = \tau_1$ are obtained by matching to Phase 2. In particular, we have $p_x(t) = p_x(\tau_1)e^{-\theta(\tau_1-t)} + \frac{h}{\theta}(1 - e^{-\theta(\tau_1-t)})$. and $p_b(t) = e^{\int_t^{\tau_1} \beta(s) ds} p_b(\tau_1) + \int_t^{\tau_1} e^{\int_t^s \beta(u) du} [h + (\mu - \theta)p_x(s)] ds$.

Since $\frac{\partial \mathcal{H}_c}{\partial g} = \frac{\partial \mathcal{H}_u}{\partial g} = \lambda_2(t)(p_x(t) - l)$, the pointwise Hamiltonian minimizer is

$$g^*(t) = \mathbf{1}\{p_x(t) \leq l\} \quad \text{for a.e. } t \in [0, \tau_2]. \quad (20)$$

Since $H_t^e = f^{-1}(p_x(t))$ by definition, the threshold rule (20) is equivalently $g^*(t) = \mathbf{1}\{f(H_t^e) \leq l\}$. For Phase 3, i.e., $t \in I_2$, note that $H_t^e = f^{-1}(p_x(t)) = \tau_2 - t$. For Phase 1, i.e., $t \in I_1$, we have $H_t^e = f^{-1}(p_x(t)) = (\tau_1 - t) - \frac{1}{\theta} \log(1 - \frac{\theta}{h} p_1(\tau_1)) = (\tau_1 - t) + f^{-1}(p_1(\tau_1)) = (\tau_1 - t) + C_1$.

Overall, the state trajectory under the threshold rule $g^*(t) = \mathbf{1}\{p_x(t) \leq l\} = \mathbf{1}\{f(H_t^e) \leq l\}$ is feasible for (5). The phasewise costate (p_x, p_b) satisfies the adjoint equations in each phase, matches continuously at phase boundaries, and satisfies transversality. Moreover, the control minimizes the phasewise Hamiltonian pointwise by (20). By the verification theorem (applied phase by phase on the smooth regimes and patched through the switching times), this control is optimal for (5). $\square$

A.3. Additional Two-Shock Analysis

In this section, we provide the proofs of Lemma 1 and Proposition 1.

Proof of Lemma 1 Let $h(t) := g(t) - g^*(t)$, $g^\varepsilon(t) := g^*(t) + \varepsilon h(t)$, $\varepsilon \in [0, 1]$. Let $(x^\varepsilon, b^\varepsilon, z^\varepsilon)$ denote the corresponding state trajectory, and define the first-order variations $\delta x(t) := \frac{d}{d\varepsilon} x^\varepsilon(t)|_{\varepsilon=0}$, $\delta b(t) := \frac{d}{d\varepsilon} b^\varepsilon(t)|_{\varepsilon=0}$, $\delta z(t) := \frac{d}{d\varepsilon} z^\varepsilon(t)|_{\varepsilon=0}$.

By the first-order expansion of the augmented functional around the optimal trajectory, we have $DJ(g^*; h) = \int_0^{\tau_2} [h(\delta x(t) - \delta z(t)) - l\lambda_2(t)h(t)] dt + \int_0^{\tau_2} p_x(t) [\lambda_2(t)h(t) - \mu\delta z(t) - \theta(\delta x(t) - \delta z(t)) - \delta \dot{x}(t)] dt + \int_0^{\tau_2} p_b(t) [(1-p)\mu\delta z(t) - \nu(t)\delta b(t) - \delta \dot{b}(t)] dt - \int_0^{\tau_2} \eta_1(t)(-\delta b(t) - \delta z(t)) dt - \int_0^{\tau_2} \eta_2(t)(\delta x(t) - \delta z(t)) dt$. By rearranging the equation and integration by part, we have $DJ(g^*; h) = \int_0^{\tau_2} \lambda_2(t)(p_x(t) - l)h(t) dt + \int_0^{\tau_2} [\dot{p}_x(t) + h(t) - \theta p_x(t) - \eta_2(t)] \delta x(t) dt + \int_0^{\tau_2} [\dot{p}_b(t) - \nu(t)p_b(t) + \eta_1(t)] \delta b(t) dt + \int_0^{\tau_2} [-h(t) - (\mu - \theta)p_x(t) + (1-p)\mu p_b(t) + \eta_1(t) + \eta_2(t)] \delta z(t) dt - [p_x(t)\delta x(t) + p_b(t)\delta b(t)]_0^{\tau_2}$. Because (p_x, p_b) satisfies the adjoint equations and terminal conditions, the terms involving δx and δb vanishes. The stationarity and complementarity conditions ensure that the terms involving δz also vanish. Therefore, $DJ(g^*; h) = \int_0^{\tau_2} \lambda_2(t)(p_x(t) - l)h(t) dt$. $\square$

Before we prove Proposition 1, we first prove an auxiliary lemma.

LEMMA 2. Under the condition of Theorem 2, $p_x(t)$ is strictly increasing on $[\tau_1, \sigma_1]$.

Proof The proof built on the proof of Theorem 2. We first note from (17) that since $\mu > \theta$ and $p_x(s) \geq 0$, $p_b(t) \geq 0$. We next show that

$$(1-p)p_b(\sigma_1) < p_x(\sigma_1). \quad (21)$$

Define $D(\Delta) := p_x(\tau_2 - \Delta) - (1 - p)p_b(\tau_2 - \Delta)$ for $\Delta = \tau_2 - t \geq 0$. By transversality, $D(0) = 0$. Differentiating with respect to Δ : $D'(\Delta) = -\dot{p}_x(\tau_2 - \Delta) + (1 - p)\dot{p}_b(\tau_2 - \Delta) = h - \theta p_x(\tau_2 - \Delta) - (1 - p)h - (1 - p)(\mu - \theta)p_x(\tau_2 - \Delta) + (1 - p)\beta(\tau_2 - \Delta)p_b(\tau_2 - \Delta)$. Rearranging and substituting $(1 - p)p_b = p_x - D$: $D'(\Delta) = ph - p_x(\tau_2 - \Delta)[p\theta + (1 - p)\mu] + \beta(\tau_2 - \Delta)p_x(\tau_2 - \Delta) - \beta(\tau_2 - \Delta)D(\Delta) = ph + (\nu(\tau_2 - \Delta) - p\theta)p_x(\tau_2 - \Delta) - \beta(\tau_2 - \Delta)D(\Delta)$. Substituting $p_x(\tau_2 - \Delta) = \frac{h}{\theta}(1 - e^{-\theta\Delta})$ yields the linear ODE:

$$D'(\Delta) + \beta(\tau_2 - \Delta)D(\Delta) = S(\Delta), \quad (22)$$

where the source term is $S(\Delta) = ph + (\nu(\tau_2 - \Delta) - p\theta)\frac{h}{\theta}(1 - e^{-\theta\Delta}) = \frac{h}{\theta}[\nu(\tau_2 - \Delta)(1 - e^{-\theta\Delta}) + p\theta e^{-\theta\Delta}]$. Since $\nu(t) \geq 0$, $h, \theta > 0$, and $p \in (0, 1)$, we have $S(\Delta) \geq hpe^{-\theta\Delta} > 0$ for all $\Delta \geq 0$. The solution to (22) with $D(0) = 0$ is $D(\Delta) = \int_0^\Delta S(u) \exp\left(-\int_u^\Delta \beta(\tau_2 - k) dk\right) du$. Since the integrand is strictly positive for $u > 0$, we have $D(\Delta) > 0$ for all $\Delta > 0$. In particular, at $t = \sigma_1$, we have (21).

Define $R(t) = p_x(t) - (1 - p)p_b(t)$. We first have $R(\sigma_1) > 0$ from (21). For $t \in [\tau_1, \sigma_1]$, differentiating R using (18) and (19): $\dot{R}(t) = \dot{p}_x(t) - (1 - p)\dot{p}_b(t) = \mu p_x(t) - (1 - p)\mu p_b(t) - (1 - p)\nu(t)p_b(t) = \mu R(t) - (1 - p)\nu(t)p_b(t)$. Since $\nu(t) \geq 0$ and $p_b(t) \geq 0$, we have $\dot{R}(t) \leq \mu R(t)$. Apply Gronwall's inequality backward from σ_1 , we have for any $t \in [\tau_1, \sigma_1]$: $R(t) \geq R(\sigma_1)e^{-\mu(\sigma_1 - t)} > 0$, where the last inequality uses $R(\sigma_1) > 0$ from (21). Therefore, $\dot{p}_x(t) = \mu p_x(t) - (1 - p)\mu p_b(t) = \mu R(t) > 0$. $\square$

Note that since $p_x(t)$ is strictly increasing on the G_1 , the threshold rule (20) implies that on Phase 2, i.e. $t \in G_1$, there is at most one switching time. Specifically, if $p_x(t) \leq l$ for all $t \in [\tau_1, \sigma_1]$, $g^* \equiv 1$; if $p_x(t) > l$ for all $t \in [\tau_1, \sigma_1]$, $g^* \equiv 0$; otherwise, there exists a unique $t^\dagger \in (\tau_1, \kappa_{\text{cong}})$ such that $p_x(t^\dagger) = l$. Then, $g^*(t) = 1$ on $[\tau_1, t^\dagger)$, $g^*(t) = 0$ on $(t^\dagger, \kappa_{\text{cong}})$.

Proof of Proposition 1 For $t \in I_2$, following the proof of Theorem 2, g^H and g^* agrees. Note that if $\tau_2 - \sigma_2 \leq K$, $g^H(t) = g^*(t) = 1$.

For $t \in G_1$, since $q(t) = 0$, $H_t = 0$. This implies that $g^H \equiv 1$. Since $p_x(t)$ is strictly increasing on $[\tau_1, \sigma_1]$, $g^H(t) - g^*(t) = 0$ for $t \in [\tau_1, t_A]$ and $g^H(t) - g^*(t) = 1$ for $t \in (t_A, \sigma_1)$. Thus, $t \in G_1$, $\Delta J^{\text{gap}} = \lambda_2 \int_{t_A}^{\kappa_{\text{cong}}} (p_x(t) - l) dt \geq 0$. If $\tau_2 - \sigma_2 \leq K$, $p_x(\sigma_1) \leq l$. Then, $t_A = \sigma_1$. In this case, $\Delta J^{\text{gap}} = 0$.

For $t \in I_1$, $p_x(t) = f((\tau_1 - t) + C_1)$. Let $u := \tau_1 - t$. The optimal rule switches at $u^* = K - C_1$, while the myopic rule switches at $u^H = K$. With $C_1 \leq K$, the disagreement window is $u \in (K - C_1, K)$, where $g^H = 1$ and $g^* = 0$. Thus, $\Delta J^{\text{carry}} = \lambda_2 \int_{K-C_1}^K (f(u + C_1) - f(K)) du \geq 0$. Substituting $f(x) = (h/\theta)(1 - e^{-\theta x})$ and integrating yields $\Delta J^{\text{carry}} = \lambda_2 \frac{h}{\theta} e^{-\theta K} [C_1 - \frac{1}{\theta}(1 - e^{-\theta C_1})]$. Expanding $e^{-\theta C_1}$ to second order gives $\Delta J^{\text{carry}} = \frac{\lambda_2 h}{2} e^{-\theta K} C_1^2 + O(C_1^3)$ as $C_1 \downarrow 0$.

Appendix B: Asymptotic Optimality

We focus on the single-surge setting; extensions to multiple surges are analogous but notationally heavier.

We first formalize the planning horizon T in (2). Given the initial state (x_0, b_0) , define $T(x_0, b_0)$ as follows. Consider a fluid system that operates at full capacity and accepts all transfer requests, starting from (x_0, b_0) . Its dynamics are $\dot{x}^o(t) = \lambda_1(t) + \lambda_2(t) - \mu(c(t) - b^o(t)) - \theta(x^o(t) + b^o(t) - c(t))$, $\dot{b}^o(t) = (1 - p)\mu(c(t) - b^o(t)) - \nu(t)b^o(t)$. The planning horizon is the first time (after the surge

onset κ) at which congestion clears: $T(x_0, b_0) = \inf\{t \geq \kappa : x^o(t) + b^o(t) \leq c(t) \mid x^o(0) = x_0, b^o(0) = b_0\}$. We also denote $\bar{V}^*(x_0, b_0)$ as the optimal value of the fluid control problem (5) with initial condition (x_0, b_0) .

To establish asymptotic optimality, we consider a sequence of systems indexed by n , in which arrival rates, transfer request rates, and capacity scale linearly with n , while abandonment and service rates remain fixed: $\lambda_1^n(t) = n\lambda_1(t)$, $\lambda_2^n(t) = n\lambda_2(t)$, $c^n(t) = nc(t)$. We also scale the initial state as $(X^n(0), B^n(0)) = (nx_0, nb_0)$. Let $Z^n(t) = \min\{X^n(t), c^n(t) - B^n(t)\}$ be the number in service and $G^n(t) \in \{0, 1\}$ the transfer acceptance decision. The dynamics satisfy

$$\begin{aligned} X^n(t) &= X^n(0) + A_1 \left(\int_0^t n\lambda_1(u) du \right) + A_2 \left(\int_0^t G^n(u) n\lambda_2(u) du \right) - S \left(\int_0^t \mu Z^n(u) du \right) - D \left(\int_0^t \theta Q^n(u) du \right), \\ B^n(t) &= B^n(0) + S \left(\int_0^t (1-p)\mu Z^n(u) du \right) - H \left(\int_0^t \nu(u) B^n(u) du \right), \end{aligned} \quad (23)$$

where $Q^n(u) = X^n(u) - Z^n(u)$, and as before, A_1, A_2, S, D, H are independent unit-rate Poisson processes.

Define the fluid-scaled processes $\bar{X}^n(t) = \frac{1}{n}X^n(t)$, $\bar{B}^n(t) = \frac{1}{n}B^n(t)$, $\bar{Z}^n(t) = \frac{1}{n}Z^n(t)$. For a policy π^n , the fluid-scaled cost is $\bar{V}^{n, \pi^n}(x_0, b_0) = \mathbb{E} \left[\int_0^{T(x_0, b_0)} h(\bar{X}^{n, \pi^n}(t) - \bar{Z}^{n, \pi^n}(t)) dt + l \bar{L}^{n, \pi^n}(T(x_0, b_0)) \right]$, where $\bar{L}^{n, \pi^n} = L^{n, \pi^n}(t)/n$ is the scaled cumulative number of rejected transfers.

For the n -th system, the fluid-scaled objective is $\min_{\pi^n} \bar{V}^{n, \pi^n}(x_0, b_0) := \frac{1}{n}V^{n, \pi^n}(nx_0, nb_0) = \mathbb{E} \left[\int_0^{T(nx_0, nb_0)} h(\bar{X}^{n, \pi^n}(t) - \bar{Z}^{n, \pi^n}(t)) dt + l \bar{L}^{n, \pi^n}(T(nx_0, nb_0)) \right]$, where $\bar{L}^{n, \pi^n}(t) = L^{n, \pi^n}(t)/n$.

For state (x, b) at time t , recall that $H_t(x, b)$ is the fluid time-to-clear if all future transfers are accepted (with arrival rate $\lambda_i(t)$, $i = 1, 2$, processing rate $\nu(t)$, and $c(t)$ servers). Although Theorem 1 characterizes the optimal fluid control through the associated optimality conditions, it is equivalently described by a single switching time: there exists a unique $\zeta^* > \kappa$ such that transfers are rejected for $t < \zeta^*$ and accepted for $t \geq \gamma^*$, where $\gamma^* = \inf\{t \geq \kappa : \frac{h}{\theta} [1 - \exp(-\theta H_t(x(t), b(t)))] \leq l\}$. Motivated by this equivalent switching time characterization, we define the look-ahead policy $\{\bar{\pi}^n\}_{n \geq 1}$ for the stochastic system as follows: transfer requests at time t if $t \geq \zeta^n$ and rejected otherwise, where

$$\zeta^n = \inf \left\{ t \geq \kappa : \frac{h}{\theta} [1 - \exp(-\theta H_t(\bar{X}^n(t), \bar{B}^n(t)))] \leq l \right\}, \quad (24)$$

i.e., the policy switches from rejection to acceptance once the predicted future congestion cost, computed via the fluid time-to-clear from the current state, falls below the blocking cost l .

Our main asymptotic optimality result is stated in the following theorem.

THEOREM 4. *Under Assumption 1, for any sequence of admissible controls $\{\pi^n\}_{n \geq 1}$,*

$$\liminf_{n \rightarrow \infty} \bar{V}^{n, \pi^n}(x_0, b_0) \geq \bar{V}^*(x_0, b_0). \quad (25)$$

Moreover, for the sequence of systems operating under the look-ahead policy $\{\bar{\pi}^n\}_{n \geq 1}$,

$$\lim_{n \rightarrow \infty} \bar{V}^{n, \bar{\pi}^n}(x_0, b_0) = \bar{V}^*(x_0, b_0). \quad (26)$$

B.1. Proof of Theorem 4

Throughout, all processes are defined on a common probability space supporting the independent unit-rate Poisson processes driving the system.

We will use a functional strong law of large numbers (FSLLN) for Poisson-driven systems with state-dependent Lipschitz rates (e.g., Mandelbaum et al. 1998): if the rates are locally bounded, Lipschitz in the fluid-scaled state, and of linear growth, then the fluid-scaled state process converges almost surely and uniformly on compact time intervals to the unique solution of the corresponding deterministic integral equation. We will apply this result on time intervals where the acceptance rule is constant (all reject or all accept).

Proof of the lower bound (25). Fix any sequence of admissible controls $\{\pi^n\}_{n \geq 1}$.

Taking expectations for (23) and using $\mathbb{E}[N(\Lambda)] = \mathbb{E}[\Lambda]$ for any unit-rate Poisson process N and any nonnegative (possibly random) time change Λ , we obtain

$$\begin{aligned}\mathbb{E}[X^n(t)] &= X^n(0) + \int_0^t n\lambda_1(u) du + \int_0^t n\lambda_2(u) \mathbb{E}[G^n(u)] du - \int_0^t \mu \mathbb{E}[Z^n(u)] du - \int_0^t \theta \mathbb{E}[Q^n(u)] du, \\ \mathbb{E}[B^n(t)] &= B^n(0) + \int_0^t (1-p)\mu \mathbb{E}[Z^n(u)] du - \int_0^t \nu(u) \mathbb{E}[B^n(u)] du.\end{aligned}\tag{27}$$

Divide by n and define $\bar{x}^n(t) := \mathbb{E}[\bar{X}^n(t)]$, $\bar{b}^n(t) := \mathbb{E}[\bar{B}^n(t)]$, $\bar{z}^n(t) := \mathbb{E}[\bar{Z}^n(t)]$, $\bar{g}^n(t) := \mathbb{E}[G^n(t)] \in [0, 1]$. Using $Q^n = X^n - Z^n$, (27) become $\bar{x}^n(t) = x_0 + \int_0^t (\lambda_1(u) + \lambda_2(u)\bar{g}^n(u) - \mu\bar{z}^n(u) - \theta(\bar{x}^n(u) - \bar{z}^n(u))) du$, $\bar{b}^n(t) = b_0 + \int_0^t ((1-p)\mu\bar{z}^n(u) - \nu(u)\bar{b}^n(u)) du$. Moreover, feasibility implies, for each t , $0 \leq \bar{Z}^n(t) = \frac{1}{n} \min\{X^n(t), nc(t) - B^n(t)\} \leq \min\{\bar{X}^n(t), c(t) - \bar{B}^n(t)\}$. Taking expectations and using Jensen's inequality for the concave map $\min\{\cdot, \cdot\}$ gives $0 \leq \bar{z}^n(t) \leq \mathbb{E}[\min\{\bar{X}^n(t), c(t) - \bar{B}^n(t)\}] \leq \min\{\bar{x}^n(t), c(t) - \bar{b}^n(t)\}$. Therefore, $(\bar{x}^n, \bar{b}^n)$ together with controls $(\bar{z}^n, \bar{g}^n)$ is an admissible solution of the (relaxed) fluid dynamics.

Next, consider the n -th fluid-scaled cost on $[0, T]$: $\bar{V}^{n, \pi^n}(x_0, b_0) = \mathbb{E}[\int_0^T h(\bar{X}^n(t) - \bar{Z}^n(t)) dt + \int_0^T l\lambda_2(t)(1 - G^n(t)) dt]$, where we used the fact that the scaled cumulative rejection count satisfies $\mathbb{E}[\bar{L}^n(T)] = \int_0^T \lambda_2(t) \mathbb{E}[1 - G^n(t)] dt$.

Using Jensen's inequality and linearity of expectation, $\bar{V}^{n, \pi^n}(x_0, b_0) \geq \int_0^T h(\mathbb{E}[\bar{X}^n(t) - \bar{Z}^n(t)]) dt + \int_0^T l\lambda_2(t)(1 - \mathbb{E}[G^n(t)]) dt = \int_0^T [h(\bar{x}^n(t) - \bar{z}^n(t)) + l\lambda_2(t)(1 - \bar{r}^n(t))] dt$. Since $(\bar{x}^n, \bar{b}^n, \bar{z}^n, \bar{g}^n)$ is feasible fluid solution and $\bar{V}^*(x_0, b_0)$ is the infimum over all feasible controls, we have $\int_0^T [h(\bar{x}^n(t) - \bar{z}^n(t)) + l\lambda_2(t)(1 - \bar{r}^n(t))] dt \geq \bar{V}^*(x_0, b_0)$. Thus, for every n , $\bar{V}^{n, \pi^n}(x_0, b_0) \geq \bar{V}^*(x_0, b_0)$, and taking $\liminf$ yields (25).

Proof of (26). Now consider the look-ahead policy $\{\bar{\pi}^n\}$ defined by (24). Let $(x^*(t), b^*(t), z^*(t), r^*(t))$ be the optimal fluid solution, where $r^*(t) = \mathbf{1}\{t \geq \zeta^*\}$ with ζ^* the unique fluid switching time. We prove that under $\bar{\pi}^n$,

$$\sup_{0 \leq t \leq T} |(\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t))| \xrightarrow[n \rightarrow \infty]{a.s.} 0, \tag{28}$$

and then show convergence of costs.

Step 1: FSLLN on intervals where the acceptance decision is constant. For any deterministic $s \in [0, T]$, on an interval $[0, s]$ where $G^n(t)$ is fixed equal to 0, the arrival rates and departure rates of (X^n, B^n) can be written as Lipschitz functions of the fluid-scaled state $(\bar{X}^n, \bar{B}^n)$ with linear growth, as $\bar{Z}^n(t) = \min\{\bar{X}^n(t), c(t) - \bar{B}^n(t)\}$ is Lipschitz in $(\bar{X}^n, \bar{B}^n)$ and all primitive rates are locally bounded. FSLLN applies on such intervals, yielding almost sure u.o.c. convergence of $(\bar{X}^n, \bar{B}^n)$ to the corresponding deterministic fluid trajectory driven by the same fixed reject decision.

Step 2: Convergence of the switching times $\zeta^n \rightarrow \zeta^$.* Define the fluid look-ahead statistic $\Psi(t; x, b) := \frac{h}{\theta} \left(1 - e^{-\theta H_t(x, b)}\right)$, and its stochastic analog $\Psi^n(t) := \frac{h}{\theta} \left(1 - e^{-\theta H_t(\bar{X}^n(t), \bar{B}^n(t))}\right)$. Recall that the switching time is defined as first passage time: $\zeta^* = \inf\{t \geq \kappa : \Psi(t) \leq l\}$, $\zeta^n = \inf\{t \geq \kappa : \Psi^n(t) \leq l\}$.

Fix $\varepsilon > 0$. We rely on the transversality of the fluid crossing. Under Assumption 1, $\Psi(t)$ is continuous and crosses l strictly at ζ^* . Thus, there exists $\delta > 0$ such that:

$$\inf_{t \in [\kappa, \zeta^* - \varepsilon]} \Psi(t) \geq l + \delta. \quad (29)$$

Consider the interval $[0, \zeta^* - \varepsilon]$. On this interval, the optimal fluid control is constant: $r^*(t) = 0$ (Reject). Let $(\tilde{x}(t), \tilde{b}(t))$ be the deterministic fluid trajectory strictly following the Reject policy. By uniqueness of the fluid ODE solutions, on this specific interval, $(x^*(t), b^*(t)) \equiv (\tilde{x}(t), \tilde{b}(t))$. By Step 1 (FSLLN with fixed policy), the stochastic system driven by the Reject policy, denoted by $(\bar{X}^n, \bar{B}^n)$, converges u.o.c. to $(\tilde{x}, \tilde{b})$. By the continuity of the mapping from state to statistic, $\sup_{0 \leq t \leq \zeta^* - \varepsilon} |\tilde{\Psi}^n(t) - \Psi(t)| \xrightarrow[n \rightarrow \infty]{a.s.} 0$, where $\tilde{\Psi}^n$ is the statistic associated with the “Reject-only” trajectory. Combining this with (29), for sufficiently large n , almost surely: $\tilde{\Psi}^n(t) > l$ for all $t \in [\kappa, \zeta^* - \varepsilon]$. This implies that the look-ahead policy, which chooses Reject when $\Psi^n(t) > l$, will indeed choose Reject throughout $[\kappa, \zeta^* - \varepsilon]$. Consequently, the actual process matches the “Reject-only” process ($\bar{X}^n(t) = \tilde{X}^n(t)$) on this interval, and no switching occurs. Thus, $\liminf_{n \rightarrow \infty} \zeta^n \geq \zeta^* - \varepsilon$ a.s.

For the other direction, we argue by contradiction. Suppose there exists $\varepsilon > 0$ such that the event $E^n := \{\zeta^n > \zeta^* + \varepsilon\}$ occurs infinitely often. On the event E^n , the stochastic policy maintains the Reject action throughout the interval $[0, \zeta^* + \varepsilon]$. Recall that $(\tilde{x}(t), \tilde{b}(t))$ denotes the deterministic fluid trajectory driven by the policy that always rejects, i.e., $g(t) \equiv 0$. Since the optimal fluid policy also rejects on $[0, \zeta^*)$, uniqueness of ODE solutions implies $(\tilde{x}(t), \tilde{b}(t)) = (x^*(t), b^*(t))$ for all $t \leq \zeta^*$. For $t > \zeta^*$, the optimal policy accepts, whereas the always-reject policy continues to reject. Continuing to reject yields a state at time $\zeta^* + \varepsilon$ no larger than switching to accept from ζ^* onward; consequently, $\Psi(\zeta^* + \varepsilon; \tilde{x}(\zeta^* + \varepsilon), \tilde{b}(\zeta^* + \varepsilon)) \leq \Psi(\zeta^* + \varepsilon; x^*(\zeta^* + \varepsilon), b^*(\zeta^* + \varepsilon))$. Moreover, by Assumption 1, the optimal trajectory satisfies $\Psi(\zeta^* + \varepsilon; x^*(\zeta^* + \varepsilon), b^*(\zeta^* + \varepsilon)) \leq l - \delta'$ for some $\delta' > 0$. Hence, $\Psi(\zeta^* + \varepsilon; \tilde{x}(\zeta^* + \varepsilon), \tilde{b}(\zeta^* + \varepsilon)) \leq l - \delta'$. By the FSLLN (Step 1), on the event E^n , the stochastic state converges to the “always Reject” fluid trajectory on $[0, \zeta^* + \varepsilon]$. Continuity of Ψ then implies $\Psi^n(\zeta^* + \varepsilon) \xrightarrow[n \rightarrow \infty]{a.s.} \Psi(\zeta^* + \varepsilon; \tilde{x}(\zeta^* + \varepsilon), \tilde{b}(\zeta^* + \varepsilon)) \leq l - \delta'$. Thus, for sufficiently large n , we have $\Psi^n(\zeta^* + \varepsilon) < l$. By definition, ζ^n is the first time the statistic drops below l . The fact that the condition is already satisfied at $\zeta^* + \varepsilon$ contradicts the assumption that $\zeta^n > \zeta^* + \varepsilon$ on E^n . Thus, $\limsup_{n \rightarrow \infty} \zeta^n \leq \zeta^* + \varepsilon$ a.s.

Since ε is arbitrary, we conclude $\zeta^n \rightarrow \zeta^*$ almost surely.

Step 3: Stitching the fluid limits across the (asymptotically matching) switching times. Define $\tau_-^n := \zeta^n \wedge \zeta^*$ and $\tau_+^n := \zeta^n \vee \zeta^*$. On $[0, \tau_-^n]$, both the stochastic policy and the fluid optimal policy reject transfers. The FSLN in Step 1 implies $\sup_{0 \leq t \leq \tau_-^n} |(\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t))| \xrightarrow[n \rightarrow \infty]{a.s.} 0$. Next, we handle the transition. By Step 2, $\tau_+^n - \tau_-^n \rightarrow 0$ a.s. Using the local boundedness of the transition rates, the state cannot change macroscopically during this vanishing interval. Thus, $(\bar{X}^n(\tau_+^n), \bar{B}^n(\tau_+^n)) \xrightarrow[n \rightarrow \infty]{a.s.} (x^*(\zeta^*), b^*(\zeta^*))$. On the remaining interval $[\tau_+^n, T]$, both policies accept transfers ($r = 1$). Let $(\hat{x}^n(t), \hat{b}^n(t))$ denote the unique solution to the Accept fluid ODE starting at time τ_+^n with initial state $(\bar{X}^n(\tau_+^n), \bar{B}^n(\tau_+^n))$. We decompose the error using the triangle inequality:

$$|(\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t))| \leq \underbrace{|(\bar{X}^n(t), \bar{B}^n(t)) - (\hat{x}^n(t), \hat{b}^n(t))|}_{(A)} + \underbrace{|(\hat{x}^n(t), \hat{b}^n(t)) - (x^*(t), b^*(t))|}_{(B)}.$$

Term (A) vanishes almost surely by applying the FSLN (as in Step 1) to the process starting at τ_+^n . Term (B) vanishes by the continuous dependence of ODE solutions on initial state. Combining these yields $\sup_{\tau_+^n \leq t \leq T} |(\bar{X}^n(t), \bar{B}^n(t)) - (x^*(t), b^*(t))| \xrightarrow[n \rightarrow \infty]{a.s.} 0$. Combining the convergence on $[0, \tau_-^n]$ and $[\tau_+^n, T]$ proves (28).

Step 4: Convergence of costs. First, the function $\bar{Z}^n(t) = \min\{\bar{X}^n(t), c(t) - \bar{B}^n(t)\}$ is continuous in the state $(\bar{X}^n, \bar{B}^n)$. Given the uniform convergence established in (28), we have $\sup_{0 \leq t \leq T} |\bar{Z}^n(t) - z^*(t)| \xrightarrow[n \rightarrow \infty]{a.s.} 0$. Next, the rejection indicator under $\bar{\pi}^n$ is $1 - G^n(t) = \mathbf{1}\{t < \zeta^n\}$, so $\int_0^T \lambda_2(t)(1 - G^n(t))dt = \int_0^{\zeta^n} \lambda_2(t)dt \xrightarrow[n \rightarrow \infty]{a.s.} \int_0^{\zeta^*} \lambda_2(t)dt$, by $\zeta^n \rightarrow \zeta^*$ and boundedness of $\lambda_2(\cdot)$.

Finally, since the fluid-scaled states are uniformly integrable (due to being dominated by the square-integrable arrival process over the finite horizon $[0, T]$), we can apply the dominated convergence theorem to exchange the limit and the expectation. Combining this with the almost sure convergence of the pathwise costs derived above, we obtain $\lim_{n \rightarrow \infty} \bar{V}^{n, \bar{\pi}^n}(x_0, b_0) = \int_0^T h(x^*(t) - z^*(t))dt + \int_0^T l \lambda_2(t)(1 - r^*(t))dt = \bar{V}^*(x_0, b_0)$, establishing (26). $\square$

Appendix C: Case Study

C.1. Data Processing

For transfer patients data, we exclude all ED-to-inpatient and inpatient-to-inpatient transfer requests, as these do not constitute ED arrivals. All ED-to-ED transfer requests are included regardless of whether the requests are accepted or declined (these account for 36% of all documented transfer requests). Together with non-transfer ED patients (external arrivals), they constitute our study cohort. We further exclude patients requesting inpatient services outside of general medicine, oncology, cardiology, surgery, neurology, or orthopedics, which accounts for 3.20% of ED patients. We include all admitted inpatients when computing hourly inpatient occupancy for each of the medical services listed above. In addition, we control for local hourly weather conditions (temperature, precipitation, snowfall, and wind speed) in all predictive analyses. Finally, to ensure the accuracy of occupancy and census measures, we exclude observations from the initial and terminal months of the data. The resulting final cohort covers April 2022 through November 2024 and con-

sists of 311,739 ED encounters. This case study was conducted with relevant Institutional Review Board approval; the need for informed consent was waived.

C.2. Estimate Processing Rate

We model the per-patient processing rate from ED boarding to inpatient admission as piecewise constant over pre-specified 8-hour intervals. Specifically, conditional on still boarding at time $t \in [t_k, t_{k+1})$, each patient completes boarding at instantaneous rate ν_k , regardless of their elapsed boarding time. For each interval k , we include all patients whose boarding periods overlap $[t_k, t_{k+1})$. Let b_i and d_i denote patient i 's boarding start and end times, respectively, and let $\mathcal{I}_k$ denote the set of patients included in interval k . Define patient i 's boarding exposure within the interval as $B_{ik} = \min(d_i, t_{k+1}) - \max(b_i, t_k)$, and let $\delta_{ik} = \mathbf{1}\{d_i \in [t_k, t_{k+1})\}$, indicating whether the patient completes boarding within the interval. Patients who remain in boarding status at the end of the interval are treated as right-censored for that interval. Conditional on the patient still boarding at $\max(b_i, t_k)$, the likelihood contribution is $\nu_k e^{-\nu_k B_{ik}}$ for a completed observation and $e^{-\nu_k B_{ik}}$ for a right-censored observation. Combining these contributions yields $L_k(\nu_k) = \prod_{i \in \mathcal{I}_k} \nu_k^{\delta_{ik}} e^{-\nu_k B_{ik}}$. Provided that total boarding exposure is positive, the maximum likelihood estimator is $\hat{\nu}_k = \frac{\sum_{i \in \mathcal{I}_k} \delta_{ik}}{\sum_{i \in \mathcal{I}_k} B_{ik}}$. Thus, the estimated processing rate equals the number of boarding completions divided by the total boarding patient-hours within the interval. We estimate the rate separately for each interval and set $\hat{\nu}(t) = \hat{\nu}_k$ for $t \in [t_k, t_{k+1})$.

Table 3 summarizes the arrival counts (per two-hour interval), boarding delays, and estimated processing rates. On average, there are 26.64 arrivals every two hours, the mean boarding delay is 17.88 hours, and the average processing rate is 0.06.

Statistic	Arrival Count (per 2 hours)	Boarding Delay (hours)	Estimated Processing Rate
Count	11700	65582	2925
Mean	26.64	17.88	0.06
Standard deviation	13.07	15.42	0.04
Min	1	0.02	0.00
25%ile	15	5.08	0.03
50%ile	27	14.78	0.06
75%ile	37	25.12	0.08
Max	67	155.03	0.30

Table 3 Summary statistics

C.3. Characterize Arrival Pattern

To characterize the temporal spacing of high-demand periods in the empirical data, we define a high-arrival interval as a two-hour interval in which the arrival count exceeds its empirical 90th percentile, and group consecutive high-arrival intervals into a single high-arrival episode. We identify 875 such episodes over the study period. The median episode lasts 2 hours, and the median gap between consecutive episodes is 18 hours. Moreover, 25.9% of episodes are followed by another high-arrival episode within 6 hours and 75.8% within 24 hours. Thus, high-arrival episodes in the empirical data are not necessarily separated by long recovery periods.

C.4. Additional Results of Case Study

In Section 6.3, we simulate the system under hospital's current policy with thresholds $\bar{b}$. Figure 9(a) summarizes the results for $\delta = 1$ and $\gamma = 0.1$. As $\bar{b}$ increases, average waiting time rises, while

the fraction of rejected transfer requests declines, reflecting the congestion–access trade-off. We next evaluate the look-ahead policy (13) over the parameter grid in (14). Figure 9(b) shows the resulting trade-off between congestion and access for $\delta = 1$ and $\gamma = 0.1$. As l/h increases, the policy places greater weight on access, rejects fewer transfer requests, and tolerates longer waiting times. Figure 9(c) presents the analogous trade-off between boarding census and transfer rejection rate. Results for all decision window sizes, $\delta \in \{1, 2, 4, 8, 12\}$ hours, and $\gamma \in \{0.0331, 0.1, 0.2\}$, are qualitatively similar and are reported in Figures 10. As discussed in Section 6.3, these patterns are less pronounced when $\gamma = 3.31\%$, as shown in the top row of Figure 10, because transfer patients constitute a smaller share of arrivals.

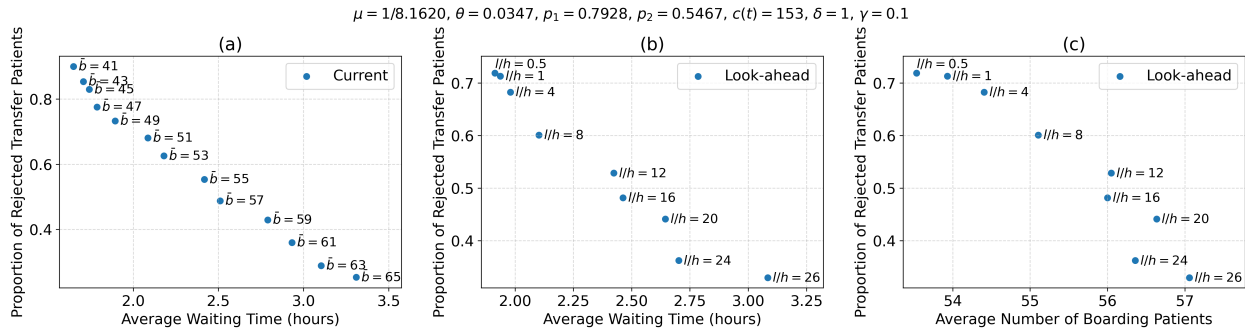


Figure 9 Simulation results under hospital’s current policy and look-ahead policy with $\delta = 1$ and $\gamma = 0.1$.

C.5. Robustness to an Expanded Forecasting Specification

In the main case study, the forecasting models use only variables that are available at each review epoch, including temporal indicators, weather variables, and lagged arrival or processing rates. This specification reflects the information set that can be used to generate forecasts in real time when implementing the look-ahead policy.

As a robustness check, we consider an expanded forecasting specification that incorporates additional contemporaneous operational information. Specifically, for arrival-rate prediction, we additionally include the ED census at the beginning of each interval. For processing-rate prediction, we further include the current boarding census and service-level inpatient occupancy measures. These variables provide a richer characterization of the system state. In this exercise, we use their realized values for the intervals being predicted; an operational implementation would require additional models to forecast their future values. Accordingly, this exercise assesses sensitivity to a richer information set.

Table 4 reports the predictive accuracy of the arrival and processing rate models. Linear regression and XGBoost achieve similar test-set performance for both prediction tasks.

Prediction Target	Dataset	Linear Regression	XGBoost
Arrival count	Train	5.61	4.23
	Test	5.69	5.67
Processing rate	Train	0.02	0.01
	Test	0.02	0.02

Table 4 RMSE values for the arrival and processing rate prediction models.

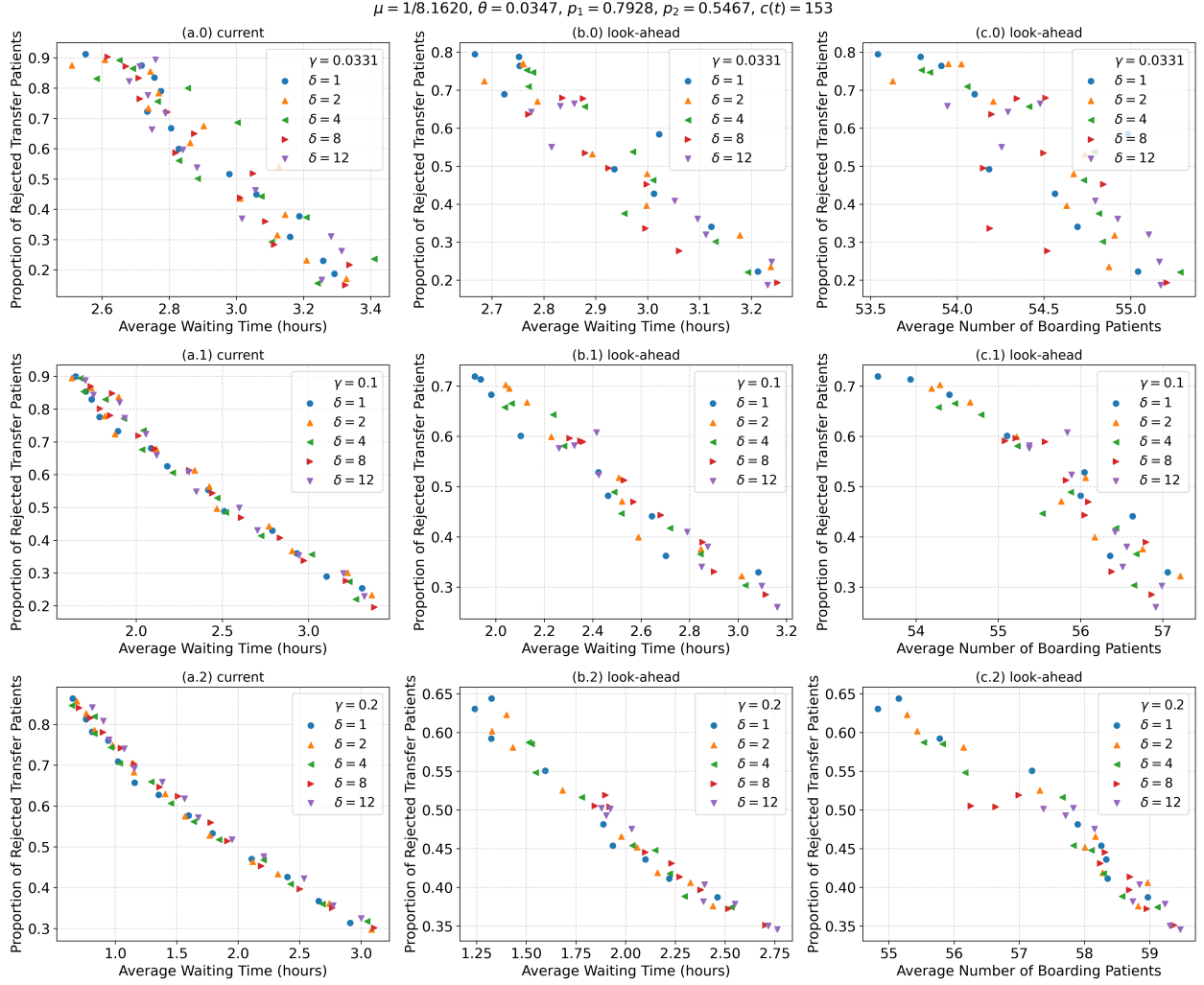


Figure 10 Trade-offs between the proportion of rejected transfers and average waiting time or boarding census.

For arrival count prediction, linear regression and XGBoost achieve similar test-set RMSEs of 5.69 and 5.67, respectively. For processing rate prediction, both models achieve a test-set RMSE of approximately 0.02. These results indicate comparable out-of-sample performance across the two models. Using these forecasting rates from the corresponding XGBoost models, we repeat the simulation study described in Section 6.3. All remaining simulation designs are unchanged.

Figure 11 compares the hospital's current policy with the look-ahead policy under the main and expanded forecasting specifications. Both look-ahead specifications achieve a more favorable congestion-access trade-off than the hospital's policy. Their performance is similar, although the expanded specification performs slightly better at some operating points, consistent with its modest improvement in predictive accuracy.

Appendix D: Additional Numerical Experiments

This section reports the numerical computation of H_t and additional numerical experiments that supplement the results presented in Section 5.

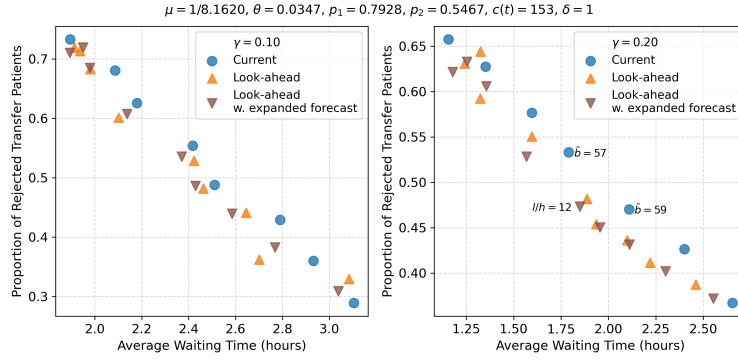


Figure 11 Comparison between the look-ahead policies and the hospital's current policy

Numerical computation of H_t . The uncontrolled system in (6) is solved numerically forward from $(\tilde{x}(t), \tilde{b}(t)) = (x, b)$ under $g(u) \equiv 1$. For intuition, under a forward-Euler discretization with step size Δ , given $(\tilde{x}_j, \tilde{b}_j)$ at time u_j , we set $\tilde{z}_j = \min\{\tilde{x}_j, c(u_j) - \tilde{b}_j\}$, $\tilde{q}_j = \tilde{x}_j - \tilde{z}_j$, and update $\tilde{x}_{j+1} = \tilde{x}_j + \Delta [\lambda(u_j) - \mu\tilde{z}_j - \theta\tilde{q}_j]$, $\tilde{b}_{j+1} = \tilde{b}_j + \Delta [(1-p)\mu\tilde{z}_j - \nu(u_j)\tilde{b}_j]$. The iterations continue until the queue clears, and the corresponding elapsed time gives $H_t(x, b)$. In our implementation, we use the `solve_ivp` routine from SciPy scientific computing library in Python with the LSODA method and adaptive time stepping rather than a fixed-step Euler scheme, together with event detection to identify the first queue-clearing time.

Benchmark policy performance. Figure 12 presents the trade-off curves of benchmark policies with $\eta \in \{0, 0.5, 1\}$ showing that, when Γ is appropriately tuned, the performance of the benchmark policy is largely insensitive to the choice of η .

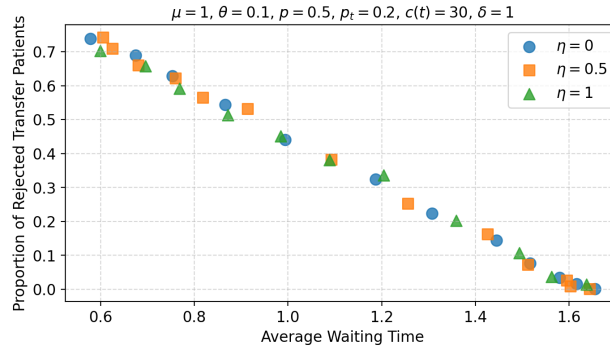


Figure 12 Trade-off under benchmark policy with different η .

Time-varying processing rate $\nu(t)$. We next consider a time-varying processing rate $\nu(t)$ with a congestion surge, where $\nu(t) = 0.7$ for $0 \leq t \leq 50$, $\nu(t) = 0.3$ for $50 < t \leq 100$, $\nu(t) = 0.7$ for $100 < t \leq T$, and $\nu(t) = 0$ otherwise. The left panel of Figure 13 illustrates the resulting profile of the processing rate $\nu(t)$. The arrival rate is set to $\lambda(t) = 15$ throughout the horizon.

To compare the performance of the benchmark and look-ahead policies, the right panel of Figure 13 depicts the trade-off between patient waiting time and the proportion of rejected transfer patients. Across both panels (corresponding to review intervals $\delta = 1$ and $\delta = 5$) the look-ahead policy consistently dominates the benchmark by achieving lower rejection rates for comparable or

shorter waiting times. These results further highlight the value of incorporating predictive information into proactive admission control decisions.

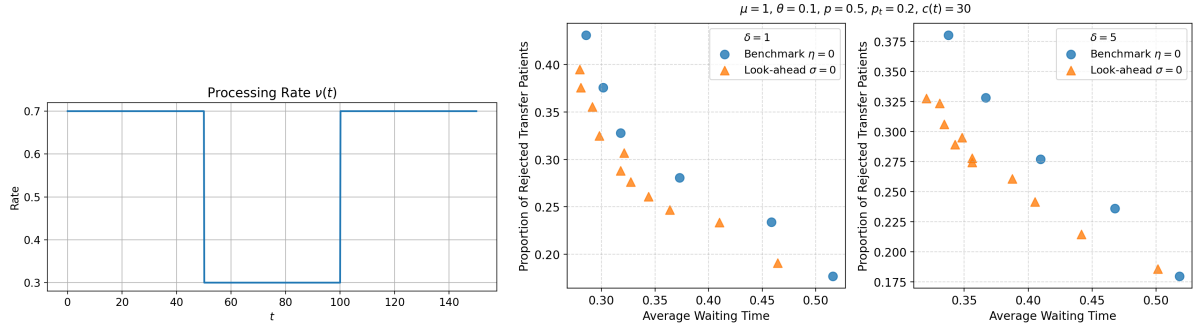


Figure 13 Comparison between look-ahead v.s. benchmark under surge processing rate.

Prediction errors. In Section 5.2, Scenarios I–V examine all combinations of a single congestion shock, two congestion shocks, $\sigma \in \{1, 2\}$, and $\delta \in \{1, 5\}$. Complete results for all configurations are available at <https://tinyurl.com/e6m37vw2>. The qualitative insights reported in Section 5.2 are consistent across all experiments.

Appendix E: Extension to Multiple Patient Classes

The main model aggregates patients receiving ED treatment into a single class. This appendix considers two patient classes that differ in their clinical and cost parameters and are served under a preemptive-priority discipline. We develop a class-specific look-ahead admission heuristic and compare its performance with a benchmark based on the fluid control.

E.1. Two-Class Model

Stochastic model. Let $\mathcal{K} = \{H, L\}$ denote the patient classes, with class H receiving preemptive priority over class L . Patients arrive through external arrivals ($i = 1$) and transfer requests ($i = 2$), with aggregate rates $\lambda_i(t)$. Let α_{ik} denote the proportion of stream i belonging to class k , where $\alpha_{iH} + \alpha_{iL} = 1$. The class-specific arrival rates are $\lambda_{ik}(t) = \alpha_{ik}\lambda_i(t)$, $i \in \{1, 2\}$ and $k \in \mathcal{K}$. Thus, external arrivals and transfer requests may have different class compositions.

A class- k patient has an exponential ED service time with rate μ_k and an exponential patience time with rate θ_k . Following ED treatment, the patient is discharged with probability p_k and requires inpatient admission with probability $1 - p_k$. Patients requiring inpatient admission enter a common boarding pool, with the time-varying processing rate $\nu(t)$, and continue to occupy ED servers until their inpatient transfers are completed.

Let $X_k(t)$, $Z_k(t)$, and $Q_k(t)$ denote the number of class- k patients who have not completed ED treatment, are receiving treatment, and are waiting, respectively, so that $X_k(t) = Z_k(t) + Q_k(t)$. Let $B(t)$ denote the number of patients boarding and define $C(t) = [c(t) - B(t)]^+$. Under work conservation and preemptive priority, $Z_H(t) = \min\{X_H(t), C(t)\}$, $Z_L(t) = \min\{X_L(t), [C(t) - Z_H(t)]^+\}$. Patients are served FCFS within each class, and boarding patients cannot be preempted.

Let $G_k(t) \in \{0, 1\}$ indicate whether a class- k transfer request arriving at time t is accepted. Let A_{ik} , S_k , D_k , and R be independent rate-one Poisson processes representing class-specific arrivals, service completions, abandonments, and inpatient transfer completions, respectively. The stochastic dynamics are $X_k(t) = X_k(0) + A_{1k} \left(\int_0^t \lambda_{1k}(u) du \right) + A_{2k} \left(\int_0^t \lambda_{2k}(u) G_k(u) du \right) - S_k \left(\int_0^t \mu_k Z_k(u) du \right) - D_k \left(\int_0^t \theta_k Q_k(u) du \right)$, $k \in \mathcal{K}$, and $B(t) = B(0) + \sum_{k \in \mathcal{K}} S_k \left(\int_0^t (1 - p_k) \mu_k Z_k(u) du \right) - R \left(\int_0^t \nu(u) B(u) du \right)$.

Fluid model and optimal control problem. Let x_k, z_k, q_k , and b denote the corresponding fluid quantities, and let $g_k(t) \in [0, 1]$ be the fraction of transfer fluid accepted. The fluid dynamics satisfy

$$\dot{x}_k(t) = \lambda_{1k}(t) + g_k(t) \lambda_{2k}(t) - \mu_k z_k(t) - \theta_k q_k(t), \quad k \in \mathcal{K}, \quad \dot{b}(t) = \sum_{k \in \mathcal{K}} (1 - p_k) \mu_k z_k(t) - \nu(t) b(t), \quad (30)$$

where $q_k(t) = x_k(t) - z_k(t)$, $z_H(t) = \min\{x_H(t), [c(t) - b(t)]^+\}$, $z_L(t) = \min\{x_L(t), [c(t) - b(t) - z_H(t)]^+\}$.

To define the congestion-episode horizon, let (x_H^o, x_L^o, b^o) denote the all-admit fluid trajectory, obtained by setting $g_H = g_L = 1$. For initial state $(\mathbf{x}_0, b_0)$, where $\mathbf{x}_0 = (x_{H,0}, x_{L,0})$, define $T(\mathbf{x}_0, b_0) = \inf\{t \geq \kappa : x_H^o(t) + x_L^o(t) + b^o(t) \leq c(t)\}$, where κ is the end of the congestion shock. This horizon is determined by the reference trajectory and is fixed across admission policies. Given class-specific holding-cost rates h_k and rejection costs l_k , the fluid-control problem is

$$\bar{V}^*(\mathbf{x}_0, b_0) = \inf_{g_H, g_L} \int_0^{T(\mathbf{x}_0, b_0)} \sum_{k \in \mathcal{K}} [h_k q_k(t) + l_k \lambda_{2k}(t) (1 - g_k(t))] dt, \quad (31)$$

subject to (30) and $g_k(t) \in [0, 1]$.

E.2. Look-Ahead Heuristic

At each review epoch t , we initialize the fluid model at the observed state and propagate it using forecasts available at that epoch, assuming that all subsequent transfer requests are accepted. Denote the resulting reference trajectory by $(\hat{x}_H^o(s | t), \hat{x}_L^o(s | t), \hat{b}^o(s | t))$, $s \geq t$. Define the predicted clearing times $\hat{H}_{H,t} = \inf\{u \geq 0 : \hat{x}_H^o(t + u | t) + \hat{b}^o(t + u | t) \leq c(t + u)\}$, $\hat{H}_{L,t} = \inf\{u \geq 0 : \hat{x}_H^o(t + u | t) + \hat{x}_L^o(t + u | t) + \hat{b}^o(t + u | t) \leq c(t + u)\}$. The first is the predicted time until sufficient capacity is available to serve all class- H patients; the second is the predicted time until sufficient capacity is available to serve all patients. Since the second condition implies the first, $\hat{H}_{H,t} \leq \hat{H}_{L,t}$. For $k \in \mathcal{K}$, define $F_k(u) = h_k \int_0^u e^{-\theta_k s} ds = \frac{h_k}{\theta_k} (1 - e^{-\theta_k u})$, if $\theta_k > 0$, and $F_k(u) = h_k u$, if $\theta_k = 0$. This function measures the cumulative holding cost of an initially unit mass of class- k fluid that remains waiting until time u unless it abandons.

We use these functions to construct the congestion indices $\hat{C}_L(t) = F_L(\hat{H}_{L,t})$, $\hat{C}_H(t) = F_H(\hat{H}_{H,t}) + e^{-\theta_H \hat{H}_{H,t}} F_L(\hat{H}_{L,t} - \hat{H}_{H,t})$. The class- L index uses the total clearing time to approximate the congestion cost associated with a low-priority admission. The class- H index has two components: a term for its own waiting cost before high-priority congestion clears, and a correction for the lower-priority service it may displace. The latter uses the interval between the two clearing times, weighted by the probability of surviving the high-priority waiting period without abandoning. This cross-class correction is a heuristic approximation rather than a general characterization of the fluid marginal

cost. It is most naturally motivated by a nested congestion episode in which high-priority waiting clears before total waiting.

At each review epoch, the heuristic accepts class- k transfer requests if their rejection cost is at least as large as the corresponding congestion index: $G_k(t) = \mathbf{1}\{\widehat{C}_k(t) \leq l_k\}$, $k \in \mathcal{K}$. The acceptance indicators remain fixed until the next review epoch.

E.3. Numerical Evaluation

We compare the two-class look-ahead heuristic with a fluid-optimization benchmark based on (31). Both policies are evaluated in the two-surge stochastic system described in Section 5, with review interval $\delta = 1$. We set $\alpha_{1H} = 0.3$, $\alpha_{2H} = 0.5$, $\mu_H = 0.6$, $\mu_L = 0.9$, $p_H = 0.6$, $p_L = 0.8$, $\theta_H = \theta_L = 0.1$. Thus, transfer requests contain a larger proportion of high-priority patients than external arrivals, and high-priority patients have longer mean treatment times and a greater probability of requiring inpatient admission. We fix $h_L = 1$ and $h_H = 2$, and vary the rejection costs over $l_L \in \{0.5, 1, 2, \dots, 9, 9.9\}$, $l_H \in \{1, 2, 4, 6, 8, 10, 12, 14, 16, 18, 19\}$. Restricting attention to $l_H \geq l_L$ yields 92 parameter combinations. For each combination, we perform 200 replications per policy.

Figure 14 reports mean total costs and 95% confidence intervals, with parameter combinations ordered by the mean cost of the fluid-optimization benchmark. The look-ahead heuristic achieves costs close to those of the benchmark across the tested settings. These results provide numerical support for using class-specific clearing-time indices in a system with heterogeneous patients and service priority.

The heuristic also requires substantially less computation. At each review epoch, it propagates a single fluid reference trajectory and performs two scalar cost comparisons. The benchmark instead solves a fluid optimization problem at every review epoch. Its computational requirements depend on the current state and the remaining planning horizon, and some updates may fail to produce a solution within the time budget. Across all 92 parameter combinations, with 200 replications per combination, and 40 parallel runs, the look-ahead experiments required approximately 4 minutes of wall-clock time, compared with approximately 27 hours for the fluid-optimization benchmark.

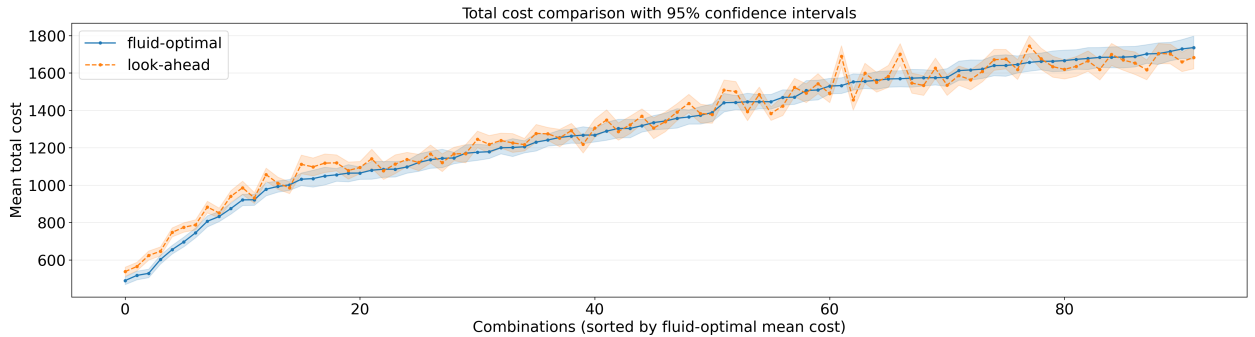


Figure 14 Two-class look-ahead policy and fluid-optimal benchmark.